Archives of Computational Methods in Engineering
https://doi.org/10.1007/s11831-025-10448-9

REVIEW

Physics-Informed Neural Networks in Materials Modeling and Design: A Review

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Received: 11 July 2025 / Revised: 29 September 2025 / Accepted: 20 October 2025
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Abstract
Physics-Informed Neural Networks are neural networks which combine data-driven deep learning and physical models by incorporating physical laws into their loss function. They are promising and flexible solvers or surrogates for problems governed by partial differential equations, including forward and inverse cases, and during the last few years they have been applied to a variety of problems in physics and engineering. In this survey, we present a comprehensive analysis of PINNs while focusing on their applications in materials science. We examine their foundational form and components, expand on enhancing strategies, and explore various aspects of their practical implementations. For materials modeling, problem cases are systematically organized based on the main scope of publications found in the literature. For this survey, problems are categorized into heat transfer related problems, including multiphysics cases, materials properties and design, materials durability, stress analysis, as well as problems related to porous media and multi-phase materials. Finally, we discuss the current challenges and limitations of PINNs, their advantages, and potential future research directions.

1 Introduction

Physics and engineering have witnessed a paradigm shift with the advent of machine learning (ML) techniques [1, 2]. During the last decades, machine learning has been thoroughly investigated and integrated into sciences in order to offer leverage and solutions in otherwise strenuous problems. A key development is physics-informed machine learning (PiML), which allows the integration of apriori knowledge and physical laws into the frameworks of ML methods for solving scientific problems involving partial differential equations (PDEs) and beyond [3]. PDEs constitute an integral part of the laws of physics, since they describe the vast majority of phenomena and problems, while their true solutions remain out of reach. Thus, the solution of PDEs remains a vital quest of the scientific community today, including the ML community, regardless of the existence and availability of well-established traditional numerical methods.

In the realm of PiML, Physics-Informed Neural Networks (PINNs) [4–6] have emerged as a promising approach that bridges the gap between data-driven deep learning, and physics-based modeling [1–3]. By incorporating physical laws and constraints into the neural network (NN) training process, PINNs offer a unique approach to many challenges. These networks embed the governing physical laws, mainly PDEs, into the loss function used for the training of the NN, while they employ the automatic differentiation method (AD) to compute the necessary derivatives. Thus, PINNs can learn both from data and physics. It is important to note that data-driven neural networks lack physical knowledge and interpretability, while they also depend on large numbers of labeled training data, often computationally expensive. PINNs can address these issues, while they are also typically mesh-free, in contrast to many traditional numerical methods used today.

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Published online: 19 November 2025

Although very recent, since the term PINNs started being popularized in 2019 [4], PINNs have been examined for a plethora of problems. A great part of relevant publications revolves around the study of foundational or pedagogical examples, but a rising number of applications involves increasingly complex cases. Furthermore, the PINNs methodology is flexible and, as a result, several improvements and strategies have been developed, such as adaptive weight schemes for the loss function [7–10], sampling strategies [11–13], boundary enforcement methods [14–16], alternative architectures [17–21], or specialized activation functions [22–24]. Today, the literature includes a diverse arsenal of PINN variants and extensions [20, 25–29], usually tailored to specific circumstances or/and employing combinations of strategies and methods.

In materials science and materials modeling/design, which are the focus of this survey, the complexity of phenomena that span multiple lengths and time scales presents particular challenges. Traditional numerical methods are reliable and widely adopted, but often struggle with the computational demands of multiscale modeling, high dimensionality, and, more importantly, inverse problems. As a result, methods of machine learning and artificial intelligence are often employed and investigated [30]. By embedding physical knowledge directly into the learning process, PINNs enable more accurate and physically consistent predictions across multiple scales [1, 4, 5].

The integration of PINNs into materials science problems and application workflows has demonstrated promising results for various examples. PINNs can infer properties of materials or model their behavior, for several cases, such as mechanical and elastic properties [31–34], photonic properties [35, 36], and electromagnetic properties [37, 38], as well as thermal properties [39–41]. Furthermore, PINNs can solve a variety of problems, such as heat transfer problems [42–47], electromagnetism problems [48], flow - transport problems [49, 50], or multiphysics problems [51, 52], including unique approaches for additive manufacturing [53, 54]. PINN frameworks have been used for several distinct challenges relevant to materials durability, such as defect analysis [55, 56], and crack propagation [57]. Some approaches focus on materials design [58, 59] or on multiscale settings [60, 61], while efforts have also been made for stress analysis related cases [62, 63].

Generally speaking, the purpose of PINNs is not to replace traditional numerical solvers, at least not at the present time frame. After all, they demonstrate different characteristics and strengths or weaknesses as tools. Their efficiency, interpretability, and flexibility, as well as their ability to be tailored to specific problems and, especially inverse cases, produce promising results and behaviors, earning a prominent place in the scientific community. Today, PINNs can act as solvers for forward and inverse settings, surrogates, or as parts of larger frameworks.

This review is two-part. We, as authors, aim to provide a comprehensive overview of PINNs regarding (i) their basic components, proposed fortification strategies and improvements, specialized variants, and (ii) their applications in materials modeling and design, by investigating a diverse collection of cases present in the literature.

Regarding the building blocks of PINNs, we present numerous methods and strategies in the literature that combat some limitations of PINNs, improve accuracy, or tailor them to the type of problem being examined. Additionally, we note that there is no universal method to classify all possible materials science problems present in the various fields of physics and engineering, especially for the cases of multiphysics and coupled phenomena. In this work, materials science cases are provided in an organized manner; starting from studies focusing on heat transfer mechanisms and problems, including multiphysics cases, to materials properties and behavior, materials design and multiscale-focused applications. These are followed by studies related to materials durability, and stress analysis, and concluded with a section dedicated to porous media and multi-phase materials. This variety is provided purposefully to demonstrate that PINNs can provide a specialized, computationally efficient, hybrid, and elegant approach to deep learning in materials science.

2 Theoretical Background

2.1 Neural Networks Fundamentals

Neural networks form the foundational basis of PINNs. They typically consist of interconnected layers of neurons that transform input features through nonlinear activations [64–67]. The most common type, relevant here, is the feed-forward neural network (FFNN / FFN or multilayer perceptron), which can be expressed as a universal function approximator [68]. In the context of this work, an FFNN (or Multilayer Perceptron) can be described as [6, 69]:

$$\hat{\mathbf{u}}_\theta(\mathbf{x}) = \mathbf{W}_L \cdot \sigma\left(\mathbf{W}_{L-1} \cdot \sigma\left(\ldots\sigma\left(\mathbf{W}_1 \cdot \mathbf{x} + \mathbf{b}_1\right)\ldots\right) + \mathbf{b}_{L-1}\right) + \mathbf{b}_L \tag{1}$$

where $\mathbf{x}$ represents input features (such as spatial coordinates), L is the number of layers, θ represents the network's parameters with $\mathbf{W}_i$ and $\mathbf{b}_i$ being the weights and biases of the i-th layer and $\sigma(\cdot)$ is a nonlinear activation function acting element-wise. Typically, NNs use supervised learning, meaning they are data-driven methods. The training process involves the minimization of the prediction error, and thus

these networks need a substantial amount of labeled training data.

Today, neural networks play a crucial role in Artificial Intelligence (AI) research and development, offering frameworks and solutions for a wide array of applications. Notable examples include Large Language Models (LLMs) [70–72], computer vision, image classification and object detection [73–75], and healthcare diagnostics [76–79].

In the context of materials science, neural networks have been applied effectively to various challenges, often under the umbrella of the PiML philosophy. For example, in the pivotal paper by Behler and Parrinello [80] a generalized neural network approach is introduced for the generation of density functional theory (DFT)-based potential energy surfaces (PESs). This approach achieves ab initio-level accuracy and can describe a wide range of chemical bonding types. The neural network potential (NNP) ANI-1 [81] is conceptually inspired by the work of Behler and Parrinello. However, it utilizes a modified version of symmetry functions and is designed to be transferrable, enhancing its suitability for broad applications in computational chemistry and materials science. In [82] SchNet is introduced as a deep learning architecture consisting of continuous filter convolutional layers. SchNet is used to model quantum interactions in molecules while respecting essential quantum-chemical constraints. The Hierarchical Deep Learning Neural Network (HiDeNN) [83] is an advanced deep learning framework, designed to address complex, computationally demanding engineering or science problems with minimal or no physical knowledge. HiDeNN is a generalized approach which has produced results for various problems; e.g. learning optimal computational meshes to capture stress concentrations, multiscale modeling, and data-driven discovery of governing parameters. A different approach is the Physics-Informed Multimodal Autoencoder (PIMA) [84]. PIMA introduces an unsupervised variational inference approach that fuses the experimental modalities of materials into fingerprints, which are grouped in physically interpretable clusters of material states it discovers. PIMA offers efficient cross-modal prediction and analysis. In addition, recently, the Differentiable Convex Optimization Layers technique [85] has been used to design architectures that learn elastoplasticity directly from stress-strain data by leveraging implicit layers that formulate the elastoplastic constitutive update as a convex optimization problem [86]. Other instances of NN applications for materials science include: Message Passing Neural Networks (MPNNs) for the prediction of quantum mechanical properties of small organic molecules [87], Crystal Graph Convolutional Neural Networks (CGCNN) that deliver both precise material property predictions (DFT-level accuracy) and detailed atomic-level chemical insights [88], ElemNet, a deep neural network model that takes elemental data (compositions) as input to predict various material properties [89], and physics-guided neural networks for atomistic modeling, which integrate physics-based analytical bond-order models with neural networks to enhance transferability in atomistic simulations [90, 91].

However, it is important to note that successful approaches are often designed around the fact that vanilla NNs face significant limitations when directly applied to materials modeling problems [92, 93], particularly those involving complex physical phenomena. Vanilla NNs are typically purely data-driven models that lack an inherent understanding of physical laws. Consequently, their predictions can be logically unreliable and show poor extrapolation performance [93]. This lack of physical interpretability, combined with the reliance on vast amounts of often expensive or unattainable labeled data, makes them poorly suited for many materials science challenges. These shortcomings particularly manifest in scenarios that require robust generalization to unseen conditions. The concept of PINN is one example of approaches that attempt to address such limitations.

2.2 PINNs Basics

2.2.1 The Standard PINN

PINNs are a methodological framework that uses neural networks combined with physical knowledge. Specifically, PINNs are NN architectures (MLP for the standard PINN), for which the training process incorporates the corresponding physical laws of the problem at hand. This is done by embedding governing laws, commonly PDEs, often in combination with other constraints or conditions, into the loss function. PINNs can be PDE solvers [4, 5], but also powerful tools for several types of engineering or materials science problems [1, 69]. PINNs are a well-known example of PiML.

PDE approximation via neural networks is a concept not limited to PINNs, and it has been a subject of study in the scientific community. A historical reference point is the work by Lagaris et al., who introduced foundational articles for the solution of ordinary and partial differential equations by utilizing deep learning [94, 95].

In the modern context, numerous studies have proposed deep learning-based PDE solvers. As will be discussed in detail, standard PINNs [4] are a collocation-based method that minimizes the loss function at a fixed set of discrete points in the problem domain. The Deep Galerkin Method (DGM) [96] is often considered familial to PINNs, since, for both, in practice, the loss functions employ a mean square error (MSE) formulation. DGM, however, differs in

philosophy, since it employs random resampling of discrete points in combination with MSE to approximate the L^2 integral formulation of the loss function, and is usually discussed in the context of high-dimensional PDEs. On the other hand, the Deep Ritz Method (DRM) [97] solves the variational (weak) form of a PDE, by minimizing energy functionals. DRM approximates these integrals with quadrature rules. A different approach to solving weak formulations of PDEs is Weak Adversarial Networks (WANs) [98], which employ primal and adversarial networks to model the solution and test function, respectively. In the case of the deep FOSLS method [99], the authors draw inspiration from least-squares methods and employ networks with loss functions based on first-order system least-squares (FOSLS) functionals. In contrast with the aforementioned single-solution approximators, Fourier Neural Operators (FNOs) [100] are neural operators that map functional parametric dependencies to solutions, so that they can learn an entire family of PDEs. DeepONets [101] are another neural operator scheme for mapping nonlinear relationships including ODEs and PDEs.

Historically, PINNs were first discussed in a two-part preprint article [102, 103] in 2017. Two years later, the combined version by Raissi et al. [4] was published and since then, it has been popular within the scientific community. PINNs can be employed for forward or inverse problems, while compared to several numerical methods, the standard PINN framework is mesh-free. For a general PDE system:

$$\begin{aligned} \mathcal{N}[\mathbf{u}] &= \mathbf{f}(\mathbf{x},t), \quad \mathbf{x}\in\Omega, \quad t\in[t_0,T] \\ \mathcal{B}[\mathbf{u}] &= \mathbf{g}(\mathbf{x},t), \quad \mathbf{x}\in\partial\Omega \\ \mathbf{u}(\mathbf{x},t_0) &= \mathbf{u}_0(\mathbf{x}), \end{aligned} \tag{2}$$

where $\mathcal{N}$ is a differential operator, $\mathbf{u}$ represents the solution $\mathbf{u}(\mathbf{x},t)$ of the PDE, and Ω is the spatial domain. The function $\mathbf{g}(\mathbf{x},t)$ describes the boundary conditions with $\mathcal{B}$ being the corresponding operator. Similarly, $\mathbf{u}_0(\mathbf{x})$ describes the initial conditions. The solution $\mathbf{u}$ can be *scalar* or *vectorial*/*complex*, w.r.t. the case examined. For scalar problems $\mathbf{u}, \mathbf{f}, \mathbf{g}$ and $\mathbf{u}_0$ reduce to u, f, g and u_0. $f(\mathbf{x},t)$, is what we refer to as the source term. If $f(\mathbf{x},t)=0$ the PDE is *homogeneous*, otherwise it is *nonhomogeneous*.

2.2.2 Loss Function Construction

For PINNs [4, 5, 104], the physical laws and constraints that describe the problem at hand are enforced through the loss function. The physics-related knowledge (e.g. material constitutive relationships, heat transfer mechanisms) is usually represented in the PDE(s) describing the system's behavior, as well as boundary or/and initial conditions (BCs, ICs). The standard PINN loss function is commonly described as the weighted sum of the loss terms [1, 2, 5, 6, 104]:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{res}}\mathcal{L}_{\text{res}} + \lambda_{\text{BC}}\mathcal{L}_{\text{BC}} + \lambda_0\mathcal{L}_0 + \lambda_{\text{data}}\mathcal{L}_{\text{data}} \tag{3}$$

where:

- $\mathcal{L}_{\text{res}}$ enforces satisfaction of the governing laws (PDEs) and represents the PDE residual
- $\mathcal{L}_{\text{BC}}$ ensures compliance with boundary conditions
- $\mathcal{L}_0$ ensures compliance with initial conditions
- $\mathcal{L}_{\text{data}}$ represents the mismatch between predictions and available data
- the λ parameters are weight parameters applied to their corresponding loss terms

2.2.2.1 Incorporation of Physical Constraints The loss terms $\mathcal{L}_{\text{res}}, \mathcal{L}_{\text{BC}}, \mathcal{L}_0$ appearing in Eq. (3) correspond to the physical laws and constraints of a specific problem. The physical constraints of the PDE are embedded through the AD of the network output with respect to its inputs, allowing the calculation of the PDE residuals.

The loss term $\mathcal{L}_{\text{res}}$ is commonly calculated by applying the mean square error formula on a number N_{res} of collocation points that have been sampled from the inner spatiotemporal domain $Q=\Omega\times[t_0,T]$ of the problem [4–6]:

$$\mathcal{L}_{\text{res}} = \frac{1}{N_{\text{res}}}\sum_{i=1}^{N_{\text{res}}} \left|\mathcal{N}[\hat{\mathbf{u}}_\theta](\mathbf{x}_i,t_i) - \mathbf{f}(\mathbf{x}_i,t_i)\right|^2 \tag{4}$$

This formulation represents the strong-form residual. MSE-based formulations are the most common for PINN applications. However, the losses can also be considered as integrals, often for theoretical analysis [5]. For example, the following Eq. (5) for $\mathcal{L}_{\text{res}}$:

$$\mathcal{L}_{\text{res}} = \int_Q \left|\mathcal{N}[\hat{\mathbf{u}}_\theta](\mathbf{x},t) - \mathbf{f}(\mathbf{x},t)\right|^2 dQ \tag{5}$$

For the boundary loss term N_{BC} the points are sampled at the boundary $\partial\Omega$ where the boundary conditions are enforced. Then, $\mathcal{L}_{\text{BC}}$ can be defined with MSE as [5]:

$$\mathcal{L}_{\text{BC}} = \frac{1}{N_{\text{BC}}}\sum_{i=1}^{N_{\text{BC}}} \left|\mathcal{B}[\hat{\mathbf{u}}_\theta](\mathbf{x}_i,t_i) - \mathbf{g}(\mathbf{x}_i,t_i)\right|^2 \tag{6}$$

Similarly, for the N_0 points sampled at the initial state t_0 to enforce the initial condition(s):

$$\mathcal{L}_0 = \frac{1}{N_0}\sum_{i=1}^{N_0} \left|\hat{\mathbf{u}}_\theta(\mathbf{x}_i,t_0)] - \mathbf{u}_0(\mathbf{x}_i)\right|^2 \tag{7}$$

In many cases, labeled data may be sparse or unavailable. After all, one of the main strengths of PINNs is their ability to accurately approximate PDE solutions without relying on labeled training data [2, 6, 105]. However, in case such information is available via experimental or observational data, or synthetic data generated by conventional methods, the data loss term $\mathcal{L}_{\text{data}}$ follows the same logic:

$$\mathcal{L}_{\text{data}} = \frac{1}{N_{\text{data}}} \sum_{i=1}^{N_{\text{data}}} \left| \hat{\mathbf{u}}_\theta(\mathbf{x}_i, t_i) - \mathbf{u}_i^*(\mathbf{x}_i, t_i) \right|^2 \tag{8}$$

The data loss component is significant for inverse problems [6].

In the literature, different representations of PDEs (or physical laws) (Eq. (2)) are used, depending on the nature of the problem. Eq. (3) is quite flexible; depending on the problem at hand, some terms may be omitted if initial or boundary conditions are unavailable or if no observed data are present [2, 5, 6]. In Raissi et al.'s work [4], the initial and boundary condition terms were merged as data loss; however, this is not as common in modern PINN applications.

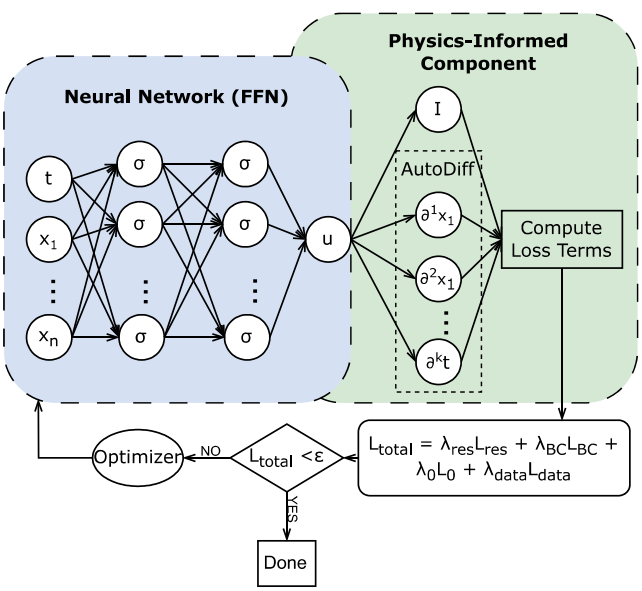

Fig. 1 A representative physics-informed neural network. The neural network, (FNN for reference), is the deep learning component of the pinn framework. It typically involves the input, hidden and output layers, the activation function(s) σ and its weights and biases for each node/ connection. The physics-informed component introduces physical laws. For PINNs, the AD method is not only used during backward propagation, but also to calculate the components of the loss function that involve derivatives of the network's output. All the loss terms are computed and contribute to the loss function. During the training, the network's weights and biases dynamically update, guided by the minimization of the loss function

2.2.3 Training Process

The training process involves minimizing the total loss function using optimization algorithms:

$$\theta^* = \arg\min_\theta \mathcal{L}_{\text{total}}(\theta) \tag{9}$$

This optimization is typically performed using gradient-based methods, where gradients are computed by automatic differentiation [5, 65]. A representative diagram of the PINN framework is in Fig. 1.

Selecting the right configuration for PINNs can greatly affect their performance and accuracy. Thus, special consideration is given to several aspects, including:

- The formulation of the loss function and weighting scheme
- The characteristics of the physical system - problem
- The generation and management of collocation and training points
- The selection of architecture,

and more. Various components of PINNs, enhancing strategies, and variations are discussed in the following section (Sect. 2.3).

2.3 PINNs Components & Strategies

In the literature there exist numerous PINNs extensions and configurations, as will be apparent in Sect. 3. A PINN configuration is often tailored to the problem at hand. In the following subsections, we discuss strategies that enhance the methodology of PINNs by improving or adapting their main components.

2.3.1 Loss Balancing and Adaptive Weighting Schemes

The loss function is one of the most critical components of the training process. The values chosen for the weight parameters λ in Eq. (3), which has a multiobjective nature [106], can greatly influence the convergence behavior and accuracy of PINNs [5, 104, 107, 108]. The use of fixed weights can be adequate for some cases, by using a trial-and-error approach to adjust them, but this can be inefficient and suboptimal [7, 109, 110]. To this end, several dynamic weight update strategies and alternatives are present in the literature. In the following paragraphs, various examples are organized thematically.

2.3.1.1 Gradient-Based Balancing Wang, Teng, and Perdikaris [7], highlight that PINNs can suffer from gradient

pathologies due to stiffness in the gradient flow dynamics. To mitigate this, the authors propose an adaptive learning-rate annealing scheme, which dynamically adjusts the loss term weights by leveraging back-propagated gradient statistics. This method updates the weights using a moving average. In [109], a similar strategy is recommended. AW-PINNs are based on adaptive weight functions, and for this approach a logarithmic mean is used to simplify the update formula. Maddu et al. [111] propose the inverse Dirichlet method for loss re-weighting, which balances task-specific objectives by equalizing the variance of their gradients. Expanding on their previous ideas, Wang et al. [8] have also introduced an NTK-based adaptive weighting strategy, which dynamically adjusts the loss term weights based on the eigenvalues of the Neural Tangent Kernel (NTK). By ensuring that different loss components converge at comparable rates, this method improves the stability and accuracy of the training.

2.3.1.2 Uncertainty-Based Weighting In [112], the authors propose Gaussian probabilistic models to define the adaptive weights present in the loss function. The recommended self-adaptive loss function method updates the weights according to maximum likelihood estimation, resulting in the self-adaptive loss balanced physics-informed neural networks (lbPINNs). The AMAW-PINN [113] employs a multitask learning approach based on homoscedastic uncertainty to adaptively adjust the weights of the loss function for each task. A different uncertainty-based allocation scheme is PINNs with weighted losses (PINN-WLs) [114] which is tailored for manufacturing systems. This method quantifies the variance of prediction errors to allocate the weights of the loss function, in order to minimize the negative impacts of data volatility and physical inaccuracies.

2.3.1.3 Attention and Residual-Based Weighting Rather than weighting entire loss terms, some methods focus on the importance of individual collocation points. These approaches often take inspiration from attention mechanisms. Anagnostopoulos et al. [9] introduce a residual-based attention (RBA) scheme, a gradient-free weighting method where the weights for the collocation points act as an attention mask dynamically scaled by the residuals themselves. Similarly, self-adaptive PINNs [10] (SA-PINNs) employ a trainable multiplicative soft attention mask, allowing the model to adaptively weight different solution regions during training. DASA-PINNs [115], employ a differentiable adversarial self-adaptive weighting scheme that automatically optimizes the point-wise loss weights of the residual loss in each training epoch. The authors adopt a bi-level optimization scheme.

2.3.1.4 Alternative Formulations and Strategies In [116], Wang et al. discuss that PINNs may violate physical causality, which is prevalent in chaotic dynamics. Thus, a weighted alternative formulation of the residual loss is proposed, where weights are determined by a causality parameter and the cumulative residual loss from previous time steps, a strategy referred to as causal training [110, 116]. A different dynamic weight update strategy, dwPINNS, that draws inspiration from the minimax algorithm to balance the weighted loss terms is introduced in [117]. Finally, van der Meer et al. [106] introduce an optimally weighted loss function using a single, fixed weight parameter. They derive its exact (but practically unknown) expression and then propose a heuristic approximation, namely magnitude normalization, that adaptively balances the terms based on the current network prediction.

2.3.2 Architectures

Today, some of the most common neural network architectures include feedforward neural networks (FNN/FFNN) - Eq. (1), convolutional neural networks (CNN), and recurrent neural networks (RNN). However, there are a multitude of architecture variants and alternatives [118, 119]. It is important to note that in order to fully describe an artificial neural network architecture, one needs to consider all its building blocks, including depth and width, activation functions, connection patterns (fully connected or sparse), or other specific elements. Additionally, the choice of architecture is problem-dependent and can vary. For PINNs applications, different configurations might be tested. Note that the term *architecture* can refer to both the high-level type (e.g., FFN or CNN) and its specific characteristics. In Table 1, the PINN architecture types in this section are organized per architecture backbone or scheme.

Feedforward networks with multiple layers are the most common architecture choice in the PINN literature. In a fully connected feedforward neural network (FFNN/FNN), each neuron is connected to every neuron in the adjacent layers [64] (see Eq. 1). In this context, the term Multilayer Perceptron (MLP) [5] is also commonly used. In the original PINNs proposal, several FFN configurations were used [4, 5]. MLP-based PINNs are those that demonstrate the mesh-free attribute.

Regardless of the prevalence of the standard MLP architecture, other approaches can also be employed. For example, pruning some of the MLP connections productively and

Table 1 Architecture types and schemes for PINNs

Architecture	PINN Variant	Description
MLP/FFN	Standard PINN [4]	Standard FFN/MLP with $\sigma(Wx+b)$ transformations.
	SPINN [120]	A sparse FFN utilizing kernel functions/networks.
	Modified MLP [7]	MLP featuring multiplicative interactions between different input dimensions and residual connections.
	ResNet Blocks [40, 121]	MLPs-based PINNs enhanced with skip connections.
	Fourier Features [122, 123]	Fourier feature embeddings to mitigate spectral bias.
CNN	PICNN [124, 125]	CNN architecture combined with Finite Volume Method.
	PhyGeoNet [17]	CNN equipped with elliptic mapping to transform between the irregular physical and the regular reference domains.
	Physics-Constrained [105]	CNN-based variant for surrogate modeling.
GNN/GCN	GCN-based [18, 126]	GCN as a prediction module combined with physics-informed loss function.
	PI-GGN [127]	GCN-based architecture combined with Galerkin-based methodology.
Equivariant	S-PINN [128]	GENNs that respect spatio-temporal parity symmetry.
	AsPINN [129]	A framework combining symmetric blocks, attention and composition to discover and enforce symmetries.
GAN	PI-GAN [19]	GAN frameworks where Generators are physically informed and constrained.
RNN	PIRNN [130]	LSTM cells ensure continuity of field variables in time.
	PhyCRNet(−s) [131]	ConvLSTM architectures with encoder-decoder scheme and time marching simulation.
	PhyLSTM^{2,3} [132]	Multi-LSTM architectures for structural metamodeling.
KAN	PIKAN [133, 134]	B-splines on edges replace activation functions on nodes.
Shallow	PIELM [135]	Physics-informed ELMs with fixed input weights for accelerated training.
	PIRBN [136]	Radial basic functions with fixed centres replace activation functions.
	PI-PINN [137]	Single hidden layer network with a polynomial interpolation-inspired strategy.
DeepONet	Physics-Informed [138, 139]	A branch-trunk dual network scheme that approximates solution operators.
Multi-network	Domain Dec. [20, 26, 140, 141]	PINNs combined with domain decomposition techniques and strategies.
	Ensembles [142, 143]	An ensemble of PINNs trains in parallel, allowing for training improvements and uncertainty quantification.
	Mixed [144]	Separate networks for primary variables and the physically related flux quantities.
	Multifidelity PINNs [21]	Four-network PINN scheme integrating multi-fidelity data.

purposefully can be advantageous [65], since shallow or sparse approaches can reduce computational costs [5]. In [120], SPINN is introduced as a sparse FNN architecture approach. SPINN leverages kernel functions or networks to produce interpretable results while using fewer parameters compared to fully connected solutions.

In [7] by Wang et al., a novel *modified MLP* architecture is introduced to further enhance PINNs. Their modification exhibits two main features: (i) it explicitly incorporates multiplicative interactions between different input dimensions, and (ii) it enhances the hidden states through residual connections. This approach reduces the stiffness of the gradient flow dynamics, while it offers a significant improvement of the network's predictive accuracy for a relatively light computational/memory overload. The modified MLP is also investigated in [116].

A different architecture modification appears in [121]. Res-PINN integrates ResNet [74] blocks with fully connected physics-informed neural networks to enhance information flow. Applied to fluid flow problems, Res-PINN demonstrates improved training stability and accuracy. In addition, it effectively solves inverse problems, even in the presence of noise. ResNet-inspired skip connections integrated in an MLP-based architecture are also seen in [40]. Alternatively, in [122, 123], the authors introduce an architecture that modifies the FNNs input layer. The key mechanism is the use of multiple parallel Fourier feature embeddings with different scaling parameters. This method is designed to combat spectral bias and significantly improves PINNs for multiscale problems.

Convolutional neural networks (CNN) [73, 145], which are primarily prevalent in computer vision [146, 147] can also be used for PINNs. CNNs are not considered mesh-free since their input data is traditionally in the form of grids or images, introducing well-defined spatial relations. The physics-informed convolutional neural network (PICNN) [124, 125] is such an approach that combines the CNN architecture with a Finite Volume Method for the approximation of the PDE residual for the loss function. It is used to simulate and predict Darcy flows. PhyGeoNet [17] is a CNN-based PINN variant that employs an elliptic mapping to transform between the irregular physical domain and the regular reference domain. It is also physics-constrained and assessed by solving parametric PDEs, demonstrating its

effectiveness over fully-connected PINNs. In addition, Zhu et al. [105] propose a physics-constrained PINN framework, specialized for surrogate modeling without labeled data, and demonstrate that the CNN-based approach is more effective than FNN in capturing multiscale features. Graph Neural Networks (GNNs) are designed for graph-like inputs, allowing a structured representation of relationships between data, which can be of great utility for certain materials science challenges and physics-guided learning [148, 149]. A specific subcategory, Graph Convolutional Networks (GCN), is conceptually a generalization of convolutional neural networks to graph-structured data. Peng et al. [18, 126] examine the prediction of natural convection, for which a GCN architecture is utilized to predict physical fields combined with the training of the PINN methodology. The physics-informed graph neural Galerkin network (PI-GGN) [127] is a discrete PINN. It combines the GCN architecture with the variational form of the PDE for training, and a piecewise polynomial basis is used to reconstruct the full-field predictions. PI-GGN can handle irregular geometries with unstructured meshes.

Equivariant neural networks enforce symmetries; meaning that if the input is transformed, the output transforms in a corresponding way. Fundamental physical laws are often equivariant, while for materials modeling, properties exhibit such behaviors w.r.t. different types of transformations. In [128], the authors introduce symmetry-preserving PINNs (S-PINNs). S-PINNs employ Group Equivariant Neural Networks (GENNs) that respect spatio-temporal parity symmetry by constructing their hidden layers using group convolutions, and are shown to significantly outperform standard PINNs for the Ablowitz–Ladik model. A more flexible and generalized approach is the adaptive symmetry-recomposition PINN (AsPINN) [129]. AsPINN's unique architecture consists of three parts: (i) hard-constrained symmetric blocks that each enforce a type of symmetry, (ii) the proposed adaptive multi-head attention mechanism that defines their importance during training, and (iii) composition modules, all trained with the physics-informed loss function. This allows AsPINN to adaptively discover and enforce the correct symmetries for a given problem, even when they are complex or not fully known in advance.

Generative Adversarial Networks (GANs) are a distinct type of architecture consisting of two networks: the Generator and the Discriminator. In GANs, the architecture is particularly unique because it also defines the learning process itself, where the two networks engage in a two-player zero-sum game, adopting an adversarial approach. Yang et al. [19] introduce PI-GANs, a physics-informed extension of generative adversarial networks, designed for solving stochastic differential equations (SDEs). PI-GANs utilize Wasserstein GANs with gradient penalty (WGAN-GP), which improve stability during training. In PI-GANs, the governing physical laws are enforced by applying differential and boundary operators to the outputs of the Generator networks, ensuring that the generated stochastic processes are physically consistent. The resulting physics-informed quantities are passed to the Discriminator, which evaluates them alongside real data during training. Multiple Discriminators are used when working with multiple groups of data. The Generators and Discriminators themselves are FNNs.

Recurrent neural networks (RNNs) have not yet been thoroughly investigated within the scope of PINNs. However, there are some cases. Physics-informed RNN (PIRNN) [130] is an RNN approach for PINNs that employs LSTM cells to ensure the continuity of field variables over time steps. PhyCRNet [131], are physics-informed convolutional-recurrent learning architectures that employ an encoder-decoder scheme and time marching simulation via autoregressive and residual connections. Specifically, they employ the convolutional long short-term memory network architecture (ConvLSTM). Another LSTM-based approach is the one in [132], where PhyLSTM^2 and PhyLSTM^3 are the proposed schemes designed for structural metamodeling.

Kolmogorov-Arnold Networks (KANs) [150] are a recent development in the field of deep learning, offering a distinct alternative to the MLP architecture. Although both schemes are fully connected architectures, KANs place learnable activation functions on edges in the form of splines, unlike the fixed activation functions on nodes present in MLPs. KANs seem to benefit from both splines' and MLPs' advantages, presenting more robustness against the curse of dimensionality, better generalization capabilities in proportion to network size, and improved interpretability. They may be a promising tool for materials modeling applications (e.g. [151]), and for physics-informed networks. Physics-informed KANs (PIKANs) have recently been compared with MLP-based PINNs and DeepONets, with their advantages and disadvantages actively being investigated [133, 134, 152].

Shallow architectures, with a single hidden layer, are also valid alternatives for PINNs. The physics-informed Extreme Learning Machine (PIEML) [135, 153] is such an example. ELMs, and PIELMs by extend, constitute minimalist approaches to neural networks. The weights of the input layer are randomly assigned and remain fixed, while the output layer's weights are determined via a least squares method. The physics-informed radial basis network (PIRBN) [136] is a different approach that utilizes the structure of a radial basis network (RBN). PIRBN employs radial basis functions (RBFs) instead of typical activation functions. For PIRBN the nodes are activated near their predefined centre which is fixed for this strategy. Another single hidden layer framework is PI-PINN [137], which

employs polynomial basis functions, including orthogonal polynomials, for approximation.

Deep operator networks (DeepONet) [101], which learn nonlinear operators, are an interesting architecture scheme and deep learning approach. It is an architecture consisting of two DNNs, one designed to encode the input space (branch network) and the other to encode the locations for the output functions (trunk network). There exist physics-informed DeepONets in the literature which fit under the PiML umbrella term, e.g. [138, 139]. Although physics-informed DeepONets blur their distinctions to PINNs, it is important to note that DeepONets, unlike PINNs which are the main focus of this work, approximate *solution operators*, enabling them to generalize across families of PDEs with varying input functions or parameters.

There are approaches, including some of the examples discussed above, that implicitly or explicitly employ multi-network strategies. A multi-network approach can involve separate neural networks or PINNs, each dedicated to solving different components of a problem (e.g. [154]). Another example is domain decomposition, where the computational domain is divided into subdomains with designated networks, as seen in cPINN and XPINN [20, 26], the multi-material oriented PINN [140], or the multi-domain PINN in [141]. Ensemble PINNs [142, 143] also deploy multiple independent networks. A different case is the mixed formulation approach proposed by Rezaei et al. [144], for which separate networks are used for the primary variable(s) and their corresponding flux quantities. Meng and Karniadakis [21], introduce multi-fidelity PINNs (MPINN), a multi-network configuration specifically designed to handle multi-fidelity data. Four interconnected networks are used; the first approximates low-fidelity data, the second and third learn the correlation between low- and high-fidelity data by separately modeling its linear and nonlinear components, and the final network enforces the governing PDE constraints to the produced multi-fidelity approximation. In general, the configuration of a multi-network and the interactions between its sub-networks depend on both the problem requirements and the specific design of the multi-network itself.

2.3.3 Activation Functions

The activation function σ, as seen in Eq. (1), generally describes the output of a neuron (node):

$$\hat{y} = \sigma(\mathbf{W}\mathbf{x} + b) \tag{10}$$

Activation functions are necessary elements of the neural network's architecture, since they enable the ability to learn complex relationships in the data. Depending on the type of activation function chosen, we can simulate different types of models in machine learning [65]. Nonlinear activation functions introduce nonlinearity to the neural network, which allows it to recognize complex patterns and approximate nonlinear functions. For deep learning implementation, some of the most widely applied activation functions are Rectified Linear Unit (ReLU), Leaky ReLU, Sigmoid, Hyperbolic Tangent (Tanh) and Softmax.

In general, the choice of activation function is vital for the efficiency and accuracy of the model [155]. Compared to standard NNs, PINNs, have additional prerequisites. Smooth, continuously differentiable functions are generally favored because they ensure the regularity of the predicted PINN solutions. Tanh, the hyperbolic tangent, is very prevalent in the scientific community [5, 110], while sinusoidal functions (e.g. $sin()$) can also be applied [156, 157]. Swish [158], a smoothed approximation of ReLU [2], is also receiving attention.

The importance of smoothness is directly tied to the core mechanism of PINNs: the embedding of PDE residuals in the loss function via AD. Many physical principles and laws are described by second-order (or higher) PDEs. To compute the residual for these PDEs, the PINN must calculate the second derivative of its output with respect to its input. Via the chain rule, this calculation requires the second derivative of the activation functions in the hidden layers to be well-defined and non-zero. Thus, the significance of (sufficiently) smooth functions is commonly highlighted in the literature, as seen in [106]. In Fig. 2 there is a visual representation of the first- and second-degree derivatives of the ReLU, versus Tanh and Swish.

Adaptive activation functions have been proposed and examined in the literature. In [22, 23], a scalable hyperparameter is introduced to the activation function, and it is trained alongside the rest of the parameters of the neural networks. These functions seem to accelerate the optimization process. For locally adaptive functions, two versions are examined; (i) layer-wise, and (ii) neuron-wise. In [156], the Adaptive Blending Units for PINNs (ABU-PINN) are presented. ABU-PINN is another approach to adaptive activation functions that combines a set of selected functions linearly, with learnable coefficients. It incorporates a gate function and adaptive slopes (as in [22, 23]). The authors thoroughly investigate the effectiveness of this method in comparison with various activation functions and adaptive strategies, demonstrating its advantages.

The self-scalable tanh (Stan) was recently developed [159], specifically to improve PINNs and combat the limitations often presented by common activation functions. Stan is smooth and non-saturating, while it also has a trainable parameter. It demonstrates improved behavior and accuracy for PINNs.

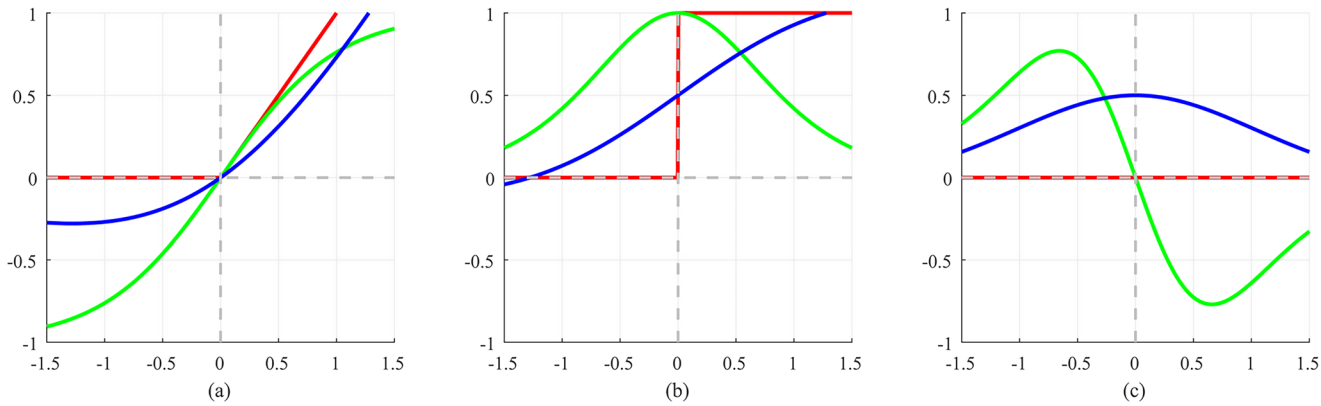

Fig. 2 Different activation functions: ReLU (red), Tanh (green) and Swish (blue), where the sub-figures demonstrate: (**a**) the original activation functions, (**b**) their first derivatives, and (**c**) their second derivatives. We observe that ReLU is highly problematic for PINNs, as its discontinuous first derivative and an undefined/zero second derivative prevent the stable computation of residuals for second-order PDEs, rendering the corresponding physics-informed loss term ill-defined or zero

Physical Activation functions (PAFS) [24] are a new concept; this new class of activation functions is inspired by the physical model of the investigated phenomena and enhances the physical constraints of PINNs. PAFs are utilized in some neurons of the neural network (either manually or in a self-adaptive manner) and can be inspired by known solutions or boundaries. PAFs enhance the extrapolation ability of PINNs, since they improve approximation for cases that do not belong to the training range. Additionally, PAFs allow for a considerable reduction of network size, without impacting accuracy.

Beyond typical NNs that are parametric models, there exist non-parametric approaches such as Gaussian Processes [160]. In the realm of PINNs, some approaches draw inspiration from this school of thought and adopt non-parametric-inspired activation functions. Some examples have been discussed in Section 2.3.2, with respect to their architecture. PIKANs [133, 134] feature learnable activation functions built from B-splines, a classic non-parametric approximator, allowing the network to learn the optimal function shape by optimizing the spline coefficients. The PIKAN framework can use most traditional approximation functions instead of B-Splines as well. The PIRBN [136] employs radial basis functions instead of typical activation functions, and the authors provide proof of convergence to Gaussian Processes when the number of nodes approaches infinity and the chosen RBF is the Gaussian function. SPINN [120] draws inspiration from non-parametric meshless methods. It also substitutes activation functions with kernel activations, which can be either simple, such as RBFs, or even small, dense MLPs named kernel networks. PI-PINNs [137] can be considered conceptually close, since the role of the activation function in the single hidden layer is replaced by power or orthogonal polynomial operations. Finally, another novel approach is by Uddin et al. [161], who introduce the use of wavelets as activation functions to improve the efficiency and performance of PINNs.

2.3.4 Optimizers

Gradient-based methods are typically used for training, with the Adam [162] and Limited-memory Broyden–Fletcher–Goldfarb–Shanno (L-BFGS), as well as L-BFGS-B, algorithms being prevalent in the literature [4–6].

In [154], a two-step training method is recommended for training; Adam is used first until a predefined stopping criterion is reached, and then the L-BFGS-B optimizer is applied for the rest of the process. The Adam + L-BFGS approach has been widely adopted (e.g. [11, 16, 37]), and it has been shown that such a combination of first- and second-order methods is superior to using a single optimizer [163].

In [164] many optimizers and their performance are investigated, and the suitability of quasi-Newton optimization algorithms, such as L-BFGS, for PINNs is also highlighted. Quasi-Newton methods also seem to improve the conditioning of the PINN loss landscape [163].

With interest in physics-informed KANs [133, 134] increasing, suitable choices for optimizers both for standard PINNs and PIKANs have been investigated in [152]. The authors extensively examine advanced quasi-Newton schemes and report that the Self-Scaled Broyden (SSBroyden) optimizer consistently outperforms other approaches, even without the utilization of common enhancing strategies.

In [165], several optimizers are examined, and the Levenberg and Marquardt (LM) algorithm showcases promising performance for the provided example cases with a small to medium number of weights. MultiAdam [166] and the Adaptive Gradient Descent Algorithm (AGDA) [167] are recently developed extensions of the Adam optimizer, designed to overcome some of Adam's limitations

for multiscale objectives, and introduce behavior tailored for PINNs, respectively. NysNewton-CG (NNCG) [163] is a novel second-order optimizer, designed to significantly improve accuracy for PINNs. It is applied after the two-step optimizer method, resulting in Adam+L-BFGS+NNCG.

A recent and novel approach is presented in [168], where meta-learners for PINNs are discussed and analyzed. These optimizers do not follow a fixed strategy like traditional ones, but instead employ a neural network in order to train their parameters and the manner in which they direct the optimization process. Adam is used as the baseline comparison, and the authors investigate it versus their learnable optimizer. The results strongly showcase that the meta-learner achieves greater accuracy at significantly lower steps.

2.3.5 Boundary/Initial Conditions

2.3.5.1 Boundary Conditions In Eq. (2), we can observe a general representation of boundary conditions. These conditions are constraints applied at the boundaries $\partial\Omega$ of a PDEs domain. They specify the behavior of the solution $\mathbf{u}(\mathbf{x}, t)$ at the edges of the domain, while they help ensure that the solution is unique and physically meaningful. They can include various types; the most common being:

- Dirichlet BCs. Dirichlet conditions, also referred to as Type I or fixed boundary conditions, are the simplest case of BCs where the values of the solution/function $\mathbf{u}(\mathbf{x}, t)$ are fixed at the boundary of the domain.

$$\mathbf{u}(\mathbf{x}, t) = \mathbf{g}(\mathbf{x}, t), \quad \mathbf{x} \in \partial\Omega \tag{11}$$

- Neumann BCs. For Neumann conditions, also referred to as Type II, the values of the normal derivative of $\mathbf{u}(\mathbf{x}, t)$ (often representing a flux or gradient) are fixed at the boundary of the domain.

$$\frac{\partial \mathbf{u}}{\partial \mathbf{n}}(\mathbf{x}, t) = \mathbf{g}(\mathbf{x}, t), \quad \mathbf{x} \in \partial\Omega \tag{12}$$

- Robin BCs. Robin conditions involve a linear combination of Dirichlet and Neumann conditions.

$$\alpha\mathbf{u}(\mathbf{x}, t) + \beta\frac{\partial \mathbf{u}}{\partial \mathbf{n}}(\mathbf{x}, t) = \mathbf{g}(\mathbf{x}, t), \tag{13}$$

  where α and β are non-zero constants.
- Mixed BCs. Mixed conditions occur when different types of boundary conditions are applied to separate regions or segments of a domain's boundary.
- Periodic BCs. For Periodic BCs (PBCs), the values of a function and its derivatives at one boundary are constrained to match those at the opposite boundary, ensuring smooth and continuous periodic behavior across the domain. Additionally, PBCs defining a small domain, called the unit cell, can represent an infinitely large system due to periodicity [169]. For PINNs and PDEs, a periodic BC is usually defined with respect to the problem's domain. For example, if $x \in [a, b]$, then a periodic boundary condition might appear as:

$$\begin{aligned} \mathbf{u}(\alpha, t) &= \mathbf{u}(b, t) \\ \frac{\partial^k \mathbf{u}}{\partial x^k}(\alpha, t) &= \frac{\partial^k \mathbf{u}}{\partial x^k}(b, t) \quad \text{for } k \geq 1 \end{aligned} \tag{14}$$

  For smooth functions g (or u) the function must belong to a differentiability class C^k, where k is the number of derivatives matched at the boundaries. For C^∞, this means that the function and all its derivatives are continuous and periodic. There can be different types of periodic boundary conditions, such as conditions with translational or rotational periodicity [170].
  Periodic conditions are not described by a function g as seen in Eq. (6). In PINNs original proposal [4, 5], Raissi et al. enforced periodic conditions directly as differences for specific problem cases (e.g. the Shrödinger example). For the boundary loss term this approach was as follows:

$$\begin{aligned} \mathcal{L}_{\text{PBC}} = \frac{1}{N_{\text{BC}}} \sum_{i=1}^{N_{\text{BC}}} \Bigg[& |\mathbf{u}(b, t_i) - \mathbf{u}(a, t_i)|^2 \\ & + \left| \frac{\partial \mathbf{u}}{\partial x}(b, t_i) - \frac{\partial \mathbf{u}}{\partial x}(a, t_i) \right|^2 \Bigg] \end{aligned} \tag{15}$$

- Cauchy BCs. The Cauchy condition specifies the function *and* the normal derivative values on the boundary, thus imposing both a Dirichlet and a Neumann condition.

$$\begin{aligned} u(\mathbf{x}, t) &= \mathbf{g_0}(\mathbf{x}, t), \quad \text{and} \\ \frac{\partial \mathbf{u}}{\partial \mathbf{n}}(\mathbf{x}, t) &= \mathbf{g_1}(\mathbf{x}) \end{aligned} \tag{16}$$

- We could also refer to *free boundary problems*, which are a type of mathematical problem that requires solving both the partial differential equation and its unknown domain Ω (thus the unknown boundary as well).

2.3.5.2 Initial Conditions Initial conditions are usually described as the initial state or snapshot of the dynamic system/PDE [171]. They describe the state of the solution $\mathbf{u}$ at the initial moment t_0, for which the most common approach

is $t_0 = 0$. When the initial conditions are Cauchy conditions, it corresponds to a *Cauchy problem*.

The nature and combination of boundary/initial conditions and whether they are well-posed or not depends on the problem at hand.

2.3.5.3 Alternative Approaches In most typical or standard PINNs, the boundary conditions are enforced in (Eq. (3)) via the corresponding loss terms. This is a *soft* boundary enforcement approach. However, *hard* boundary enforcement can be advantageous.

Through a hard boundary condition approach, no boundary loss term is used, and instead BCs are integrated into the network design. For Dirichlet/ essential conditions a common strategy is to use an ansatz that integrates the prediction of the neural network $NN(x)$ as $u_{ans} = G(x) + D(x) * NN(x)$. For example, in [14], exact or approximate distance functions (ADFs) are utilized to transform the PINNs' approximation into such a boundary-compliant solution and to avoid boundary loss terms. Their method of constructing ADFs is inspired by R-functions and generalized barycentric coordinates (mean value potential fields). This approach, the typical weighed summation loss function Eq. (3), as well as Nitsche's method [172] are applied and compared through numerical experiments for Dirichlet boundary conditions in [104]. This study involves both PINNs and VPINNs. The hPINN follows the same philosophy for hard constraints through the architecture [16].

UPINN [15] handles boundary conditions automatically in two phases realized by two networks: (i) first, a neural network learns a trial function that naturally satisfies the given boundary conditions in homogeneous form, and (ii) then the second network uses this trial function to define the constrained output and trains to minimize the loss function in either a strong or weak form.

Manavi et al. [173] examine PINNs for surrogate modeling. The authors use two neural networks to produce smooth boundary condition and distance functions, which are then incorporated into the third network, the PINN ansatz which enforces hard constraints. This approach extends the work of Berg and Nyström [174] and also investigates hybrid enforcement for mixed boundary conditions. Their framework is tested on 1D, 2D, and 3D heat conduction problems with various geometries and boundary conditions.

Dong and Ni [175] propose a method to exactly enforce periodic boundary conditions in deep neural networks through a periodic layer, which facilitates the representation of periodic functions and the integration of C^∞ and C^k conditions, making it a popular approach for many PINN frameworks (e.g., [9, 116, 176]).

2.3.6 Collocation Points

Typical PINNs utilize a number of collocation points sampled in the problem domain for the computation of residuals. For the generation of collocation points, methods such as uniform grids, uniform random sampling, Sobol sequences, and Latin Hypercube Sampling (LHS) are quite prominent. One might consider to what extent the sampling method affects the PINNs' capabilities and flexibility, while the geometry of the problem also plays a crucial role in regard to the optimal distribution of training data [5]. Furthermore, for systems with chaotic or turbulent behavior, other limitations and challenges arise [116]. In general, it is important to note that random sampling improves the generalization capabilities of the model [110] and lightens computational and memory load. Improved sampling techniques for collocation points have been investigated in the literature.

Such methods often introduce dynamic or re-sampling strategies. Wight and Zhao [13] discuss multiple strategies to enhance PINNs accuracy and efficiency, including adaptive resampling of collocation points in space, as well as (i) adaptive sampling in time, and (ii) adaptive time marching. For the first technique, a single PINN is trained in the space-time domain, starting with a small initial subset of the time domain which expands incrementally, while for the second technique, different PINNs are trained for separate time intervals, each providing the initial condition for their subsequent network. Sequential training is also recommended in [108]. In [113], the collocation points are divided into a fixed and a movable set, and an adaptive collection point movement method is proposed; it dynamically adjusts the collocation points during training, to ensure dense distribution in important areas and uniformity in regular regions.

Lu et al. [11] implemented Residual-Based Adaptive Refinement (RAR), a technique that improves the distribution of residual (collocation) points during training. This approach is inspired by the Finite Element Method. During RAR, additional points are generated for areas in which the mean PDE residual is large. The importance sampling method [177], selects training points based on a distribution proportional to a piecewise constant approximation of the loss function, improving efficiency without introducing new hyperparameters and allowing easy integration into existing PINN frameworks. The Retain-Resample-Release sampling (R3) and Causal R3 algorithms [178] are designed with the goal of mitigating propagation failures. Wu et al. [12] investigate sampling methods for fixed collocation points in depth, and also propose two adaptive methods: residual-based adaptive distribution (RAD) and residual-based

adaptive refinement with distribution (RAR-D), which is a hybridization of RAR and RAD. These two methods encompass a group of other strategies in the literature.

PINNACLE [179], is a novel algorithm that estimates training dynamics using NTK. This algorithm simultaneously optimizes all types of training points, including experimental ones, while dynamically adjusting the proportion of collocation points as training progresses.

2.3.7 Feature Engineering

Beyond the choice of loss function and architecture, a crucial aspect of sophisticated PINN modeling involves feature engineering. Feature engineering embeds prior knowledge into the network's input and output representations to create a more physically meaningful model. Although some techniques have been mentioned in different sections, their role as feature engineering warrants further discussion.

For PINNs, a foundational technique is the non-dimensionalization of the governing equations. By scaling raw physical variables using characteristic units, the network operates on dimensionless inputs and outputs. This process identifies key dimensionless numbers (e.g., Reynolds, Péclet; also see (21), (26)). These numbers describe the solution, and the choice of characteristic units directly impacts the training process [3, 110]. Dimensionless formulations allow for vanishing gradient and scale disparity mitigation, as well as improved convergence.

A different approach is feature mapping; input coordinates are transformed before being passed to the main network. Fourier feature embedding [122, 123] maps coordinates to sinusoidal features to combat the spectral bias of MLPs. A more tailored physics-guided feature engineering strategy involves adding 'pre-layers' that transform inputs based on the known analytical form of the PDE's solution, as demonstrated by Zobeiry and Humfeld [180]. This allows for a physically meaningful feature transformation that is directly correlated to the nature of the governing laws. An even more elevated example is AsPINN [129], which extracts but also dynamically calibrates several representative symmetric features from the inputs through symmetric blocks.

Governing laws can also inform the design of the network's outputs, allowing for output-oriented feature engineering. In the mixed formulation approach [144], separate networks are used to predict primary variables and their physically related flux quantities. The method allows for a drop in derivative order for calculations, improves interpretability, and enhances performance. Similar approaches can be found. T-phPINN [181] dedicates different networks to temperature and its derivative, respectively. In general, especially for coupled or complex multiphysics problems, assigning distinct physical fields to separate network outputs (e.g. [34, 154]) can be advantageous.

BCs enforced via hard constraints, such as periodic layers or ansatz-based approaches (see Sec. 2.3.5) also fall under the scope of feature engineering. Finally, when data possess an inherent relational or topological structure, transforming the training data into a grid or a graph provides the network with inductive bias. CNN- and GNN- based PINN approaches can be chosen for such cases.

2.3.8 Other PINNs Variants and Extensions

As discussed in the previous subsections, by altering the building blocks of a PINN, or/and using improvement techniques, a great number of PINNs implementations and variations are being created, improved, and investigated for improved performance or specialized application. We present some diverse examples of PINNs present in the literature:

- VPINNs. Variational PINNs (VPINNs) [25] extend the idea of PINNs by employing the variational (weak) formulation of the governing equations, resulting in a more flexible framework. VPINNs are considered a Petrov–Galerkin extension of PINNs, where the PDE is multiplied by a set of chosen test functions and integrated over the entire domain to obtain the variational form. This approach allows the problem to be solved in a less restrictive manner compared to the classical strong formulation. A prominent extension of VPINNs is variational physics-informed neural networks with domain decomposition (hp-VPINNs) [182]. The cv-PINN [183] is introduced as an improved variant, that embeds test functions and quadrature weights into convolutional filters, enabling parallelization and computational efficiency through convolutional operations.
- cPINN. Conservative PINNs (cPINNs) [26] extend PINNs for domain decomposition, specifically for solving nonlinear conservation laws. The computational domain is divided into subdomains, each assigned a separate PINN, with flux continuity enforced at the interfaces. This ensures conservation properties are preserved while improving parallelization, scalability, and efficiency.
- XPINN. Extended PINNs [20] are a generalized space-time domain decomposition framework, moreover, unlike cPINN they can be applied to arbitrary types of PDEs. XPINN provides all the cPINN advantages, while offers more room for application and simple interface conditions.
- B-PINNs. Bayesian PINNs [27] are a probabilistic extension of PINNs that incorporates Bayesian neural

networks (BNNs) to handle noisy data and quantify uncertainty. B-PINNs leverage Hamiltonian Monte Carlo (HMC), or Variational Inference (VI) to estimate the posterior distribution over the neural network parameters. This framework improves robustness against noise, prevents overfitting, and provides a principled way to estimate aleatoric uncertainty in both forward and inverse PDE problems.

- hPINN. This variant of PINNs with hard constraints (hPINN) [16], is specifically designed to address inverse problems. hPINN integrates the governing PDEs, boundary conditions, and inequality constraints into their framework. The boundary constraints are directly enforced into the network architecture, while the PDE and inequalities are enforced via the loss function. To impose PDEs and inequality constraints as hard constraints, two methods are proposed: (i) the penalty method and (ii) the augmented Lagrangian method.
- fPINN. Fractional PINNs [29] are a PINN extension intended to solve space-time fractional PDEs. A key feature of fPINNs is the hybrid approach that combines automatic differentiation for integer-order operators with numerical discretization for fractional operators, enabling effective construction of the residual in the loss function despite the challenges posed by fractional calculus. It is demonstrated for fraction advection-diffusion equations (fractional ADEs).
- nPINN. Nonlocal PINNs [184] were introduced for parameter and function inference in integral equations, particularly nonlocal models. Pang et al. propose: (i) a unified nonlocal Laplace operator that bridges classical and fractional Laplacians, providing a versatile framework; (ii) the use of nPINN to estimate parameters characterizing this operator, revealing multiple effective solutions due to the non-convex nature of the loss function; and (iii) a new nonlocal operator with spatially variable order for modeling wall-bounded turbulence, demonstrating universal behavior concerning the Reynolds number.
- bc-PINN. The backward compatible PINN (bc-PINN) [185] is a novel PINN that sequentially solves the PDE in successive time segments using a single neural network. The core concept involves retraining the same neural network for each time segment, while ensuring that it satisfies the solutions obtained for all preceding segments through a modification in the loss function. It showcases advantageous performance compared to standard PINNs, and the two enhancement techniques proposed in [185], significantly improve its accuracy and efficiency.
- Multiphysics PINNs. Multiphysics PINNs (often abbreviated as MPINNs) commonly refer to PINN frameworks extended to accommodate multiphysics problems. For example, He et al. [154] propose a MPINN for subsurface transport, using three separate DNNs, trained simultaneously, to solve the coupled Darcy flow and advection-dispersion equations and estimate necessary physical quantities. Other multiphysics PINNs employ distinct implementations to accommodate the governing physical laws.
- A-PINN. Auxiliary PINNs [186] are a PINN extension developed to handle integro-differential equations (IDEs). A-PINNs utilize a multi-output neural network architecture, where auxiliary outputs represent the integrals related to the governing equations. By adding output conditions to the governing problem and their corresponding loss terms, A-PINNs enforce constraints between the primary outputs and the auxiliary outputs in the form of physical laws.
- gPINNs [28]. Gradient-enhanced PINNs utilize gradient information of the PDE residual. The authors alter the loss function by adding a term that forces the gradient of the PDE residual to also be zero, since the residual itself is minimized towards zero. This added term improves accuracy, while it is also recommended to combine gPINNs with residual-based adaptive refinement (RAR) [11].

3 PINNs Applications in Materials Science

Materials science constitutes a vast collection of scientific challenges with a plethora of applications in engineering, mathematics, physics, manufacturing, etc. In this section, we organize materials science problems and present several corresponding PINNs approaches and implementations found in the literature. We recognize that many papers in the literature offer both methodological advances and unique applications. The focal point of this chapter is to present applications for a variety of cases, but the unique characteristics of the proposed PINNs are also noted. The papers presented in the following paragraphs are categorized according to the nature of the problem(s) being solved with their corresponding PINN frameworks, while in the previous chapter, Section 2.3, PINN enhancement strategies, methodologies, and variants are the main focus.

Depending on the problem, PINNs can be applied to either a forward (also referred to as direct) or inverse setting. A *forward problem* is one where the governing physical laws and all required inputs (e.g., boundary conditions, coefficients, source terms) are fully specified, and the goal is to compute the system's response. In contrast, an *inverse problem* seeks to infer unknown inputs by leveraging available information such as constraints or data on the system's

behavior; thus, it seeks to identify the physical system itself [2]. PINNs in inverse problems integrate data-driven learning or prior knowledge with physical constraints to estimate these unknowns. Furthermore, many problems involve multiple interacting physical processes which are modeled simultaneously, very often involving *coupled* PDEs as well. This is often the case for more realistic/ multiphysics examples (e.g. subsurface transport [154], or the rising hot bubble [51]). These can be either forward or inverse problems, depending on whether the necessary inputs are known.

In order to assess the performance and accuracy of PINNs, several strategies can be employed. When a known analytical solution is available, a PINN implementation can be tested against this ground truth to evaluate its effectiveness. However, in most real-world cases, an analytical solution may not be attainable. In such cases, PINNs can be compared with traditional numerical methods, such as Computational Fluid Dynamics (CFD) and Finite Element Methods (FEM), or against known data and benchmarks from the literature. Such approaches have been considered in the works discussed in the following paragraphs.

The works cited and discussed in the following paragraphs are summarized in Table 2 and Table 3.

3.1 Heat Transfer Problems

Heat transfer is a critical aspect of many materials science applications (such as quenching, annealing, and forging), encompassing a wide range of mechanisms, such as thermal conduction, convection, radiation, and energy transfer via phase changes. Each of these phenomena presents unique challenges and applications in various industries. The underlying physical laws governing these processes have been extensively studied, providing a solid foundation to exploit the advantages of PINNs.

In most cases, a heat transfer problem is governed by multiple coupled PDEs, depending on the physical context. Therefore, the classification of heat transfer problems does not necessarily correspond to a simple one-to-one mapping between a problem type and a single PDE. Keeping that in mind, for this work we classify the discussed PINNs publications w.r.t. their main area of focus. We use the symbol T for the temperature as seen in the following PDEs/ equations.

3.1.1 Thermal Conduction

The heat equation is the most representative PDE in the context of thermal conduction. One of the most common representations is the following:

$$\frac{\partial T}{\partial t} = \alpha \nabla^2 T + S(\mathbf{x}, t) \tag{17}$$

$\alpha = \frac{k}{\rho c_p}$ is the coefficient describing the thermal diffusivity of the medium with k, ρ, c_p being conductivity, density and specific heat respectively. $S(\mathbf{x}, t) = \frac{Q(\mathbf{x},t)}{\rho c_p}$ w.r.t Q being the volumetric heat source/sink (if applicable).

In the context of heat conduction, it is necessary to address Fourier's law, which is a fundamental principle that describes how heat flows through a material.

$$\mathbf{q} = -k \nabla T, \tag{18}$$

where $\mathbf{q}$ is the local heat flux density. Fourier's law is embedded in the heat equation.

In the case of steady-state heat transfer, the system reaches thermal equilibrium, which means that the temperature distribution does not change over time. The governing equation may be simplified to the following:

$$\nabla \cdot \mathbf{q} = Q(\mathbf{x}) \tag{19}$$

Xu et al. [42] propose PINNs with a convolutional encoder-decoder neural architecture for the prediction of temperature / heat flux fields in porous systems, enabling the inference of effective thermal conductivity. Their framework treats heat conduction as an image-to-image regression problem, where the input is a binary image of the porous microstructure, and the outputs are the predicted temperature and heat flux fields. Their investigation extends to cases of nonlinear heat conduction, which means that conductivity k in Eq. (18) varies with temperature, while no labeled data are used. In [41] PINNs are used as an inverse data-driven approach to predict conductivity in one-dimensional heat transfer. This framework is stepwise and utilizes three preliminary networks, a main one, and a reality test. It results in good approximations from significantly limited data (down to three), while the prediction of thermal conductivity values of a 18650 Li-ion battery cell is also investigated.

Papadopoulos et al. [39] use XPINNs [20] for the estimation of the Kapitza thermal resistance at the interface between distinct phases within a multi-phase composite material, demonstrating the potential of XPINNs to provide accurate and computationally efficient solutions for inverse heat transfer problems using limited internal temperature measurements. Jiang et al. [143] address space-dependent inverse heat conduction problems, which are often ill-posed due to noisy or limited data. Their proposed Ensemble PINNs employ deep ensemble learning for uncertainty quantification, enhanced through adversarial training and adaptive active sampling. Comprehensive numerical

Table 2 PINNs applications in materials modeling and design (Part A)

Application Focus	Short Description	Reference
Thermal Conduction	Convolutional Encoder-Decoder scheme for linear/nonlinear heat conduction in porous media.	Xu et al. [42]
	Inverse, data-driven strategy for thermal conductivity, also tested for 18650 Li-ion battery cells.	Jo et al. [41]
	XPINNs estimate Kapitza thermal resistance for multi-phase composite materials.	Papadopoulos et al. [39]
	Ensemble PINNs for space-dependent inverse heat conduction problems.	Jiang et al. [143]
	Tailored PINN enhancements for heat conduction for forward and inverse analysis.	He et al. [40]
	Multi-domain PINN for steady-state heat conduction in multilayer media. Displays accuracy for the thin film at nanoscale.	Zhang et al. [141]
	T-phPINN, predicting temperature and its derivative for 2d non-Fourier heat conduction.	Zheng et al. [181]
	PINNs with feature-aware pre-layers for heat conduction with convective boundaries.	Zobeiry, Humfeld [180]
Thermal Convection	Examination of PINNs for ill-posed forced or mixed convection scenarios.	Cai et al. [47]
Stefan (Benchmarks)	Dual PINNs to model solution and moving boundary for forward and inverse cases.	Wang et al. [45]
	Dual scheme enhanced with log-cosh based loss formulation.	Li et al. [187]
	PINNs solve the Stefan enthalpy formulation, combined with enhancing strategies.	Madir et al. [188]
	Dual PIELMs outperform dual standard PINNs.	Ren et al. [153]
Stefan (Complex)	PINNs enhanced with adaptive collocation method, transfer learning and neural architecture search model melting Sisko fluids.	Guo et al. [43]
	Enthalpy formulated PINNs model directional freeze-casting solidification in biomaterial scaffolds.	Rouhollahi et al. [189]
Radiative Heat Transfer	PINN investigation for radiative heat transfer for multiple, diverse, inverse and forward cases.	Mishra, Molinaro [190]
	A-PINNs solve inverse and coupled integro-differential radiative heat transfer cases.	Riganti, Dal Negro [46]
	PINNs for RTE with graded index, a complex high dimensional setting.	Murari, Sundar [44]
Multiphysics cases	Heat transfer in power electronics; ANSYS (electronic chips) and NVIDIA (heat sink designs).	Cai et al. [47]
	PINNs compared with clique polynomial method for radiative heat transfer in a concave porous fin under LTNE conditions.	Chandan et al. [191]
	Heat transfer with two-phase flows via multi-network PINNs for rising bubble cases.	Jalili et al. [51]
	Dual, disconnected scheme to model thermochemical evolution of composite material undergoing cure for single and bi-material settings.	Amini Niaki et al. [192]
	Multiphysics PINN models thermal runaway in lithium-ion batteries.	Kim et al. [52]
	PINN as a component of a multiscale framework predicts thermal conductivity for Polyurethane-Phase Change Materials foam composite.	Liu et al. [193]
	Tailored PINNs for thermo-hydro-mechanical coupled problems.	Amini et al. [194]
Additive Manufacturing	PINNs model metal AM processes: aluminum solidification and the first challenge of 2018 NIST AM-Bench.	Zhu et al. [53]
	PINNs for 3D temperature field prediction during direct energy deposition. Tested for settings of Inconel 718 and 316 L stainless steel.	Xie et al. [54]
	PINN combined with online learning for heat predictions during metal AM.	Sajadi et al. [195]
	PINNs improve physical models of the thermal behavior of Stereolithography process during cooling.	Tod et al. [196]
Mechanical Properties	COMM-PINN for path-dependent materials modeling, solves constitutive laws, encompassing yield conditions, evolution laws, and KKT conditions.	Rezaei et al. [197]
	Multinetwork PINN for coefficient estimation in the wave equation to characterize complex materials.	Lee, Popovics [32]
	RheologyNet framework estimates thixotropic properties of cementitious materials.	Zhang et al. [33]
Elastic Properties	PINN for linear elasticity and nonlinear von Mises elastoplasticity for inverse cases and surrogate modeling.	Haghighat et al. [34]
	PINNs enhanced with classical solution-based strategies for problem cases of biharmonic elasticity.	Vahab et al. [198]
	Dual PINN scheme identifies nonhomogeneous materials for the elasticity imaging problem in bio-medical settings.	Zhang et al. [31]
	PINN enhancements for calibrating elastic models from full-field displacement data in realistic settings.	Anton, Wessels [199]

Table 3 PINNs applications in materials modeling and design (Part B)

Application Focus	Short Description	Reference
Optical and Photonic Properties	PINNs retrieve optical properties in metamaterials and nano-optics cases from synthetic scattering data.	Chen et al. [35]
	PINNs obtain complex optical parameters of nanostructured environments in near-field microscopy settings.	Chen, Dal Negro [36]
	PINNs embedded with parametric launch power infer optical fiber parameters from the nonlinear Schrödinger equation.	Jiang et al. [200]
	PINNs estimate system parameters of fiber amplifiers and enhance monitoring.	Wang et al. [201]
Electromagnetic Properties	Feasibility study of PINNs for forward problems in electromagnetics.	Khan et al. [48]
	PINNs with frequency scaling for electromagnetic inverse scattering problems.	Hu et al. [37]
	PINNs and Deep Ritz Method for magnetostatic and micromagnetic problems.	Kovacs et al. [38]
	Multi-network PINN framework models the electromagnetic response of a permanent magnet synchronous motor.	Son et al. [202]
Materials Design (Optical/Photonic)	MscalePINNs for the inverse design of finite-size photonic materials with stealthy hyperuniform disordered geometries.	Riganti et al. [58]
Materials Design (Electromagnetic)	PINNs for electromagnetic metamaterials design, focused on cloaking cases.	Fang, Zhan [203]
	PINN defined by a hypernetwork act as real-time field solver for materials design in magnetostatic settings.	Baldan et al. [59]
Multiscale Materials Modeling	PINNs with domain decomposition accelerate and enhance EMsFEM.	Wu et al. [60]
	NH-PINN for solving multiscale equations through PINN-based homogenization.	Leung et al. [61]
	CPINNTO: a framework integrating PINN variants for topology optimization, with multiscale-oriented adaptation.	Jeong et al. [204]
Defect Analysis	PINNs as a nondestructive quantification method to locate surface breaking cracks in metal plates.	Shukla et al. [55]
	PINNs retrieve characteristics of voids/inclusions in matrix materials.	Zhang et al. [205]
	PINNs produce nonuniform backgrounds for heat diffusion, used to highlight defects in carbon fiber-reinforced polymer in forward and inverse settings.	Lim et al. [56]
Fracture and Crack Propagation Analysis	Enhanced PINN minimizes variational energy to incrementally simulate fracture propagation.	Goswami et al. [57]
	Variational energy-based PINNs combined with PointNet-based adaptive refinement for crack propagation in industrial settings.	Tu et al. [206]
	PINN enhanced with crack-tip enrichment functions models 2D in-plane crack analysis.	Gu et a. [207]
Fatigue Life	M-integral fatigue-inspired PINNs (FP-M and FP-P) predict the lifetime of multi-defect materials under cyclic loading.	Dong et al. [208]
	PINNs integrate a novel linear elastic fracture mechanics model to estimate fatigue for additively manufactured materials with defects.	Salvati et al. [209]
Stress Analysis	PINN and cPINN approaches predict displacement, stress, and energy fields in heterogeneous micro-structural unit cells; applied to a wood-plastic composite case.	Henkes et al. [62]
	PINNS model diffusion-induced stress in an elastic sphere for a lithium-ion battery setting.	Xue et al. [63]
Porous Media and Multi-phase Materials	Dual PINNs model hydraulic conductivity for saturated or unsaturated subsurface Darcy flows in porous media.	Tartakovsky et al. [210]
	Three-network multiphysics PINN models subsurface flow and transport for a steady-state advection–dispersion problem.	He et al. [154]
	PINNs investigated for the 1D Buckley-Leverett problem for two-phase displacement.	Almajid,Abu-Al-Saud [211]
	Multi-network PINNs model resin flow in fibre textiles for realistic scenarios.	Chen et al. [49]
	PICNNs integrate finite volume method to simulate single- or two-phase Darcy flows and incrementally predict the 2D pressure field of reservoirs.	Zhang et al. [124, 125]
	Enriched PINNs, with a specialized architecture that integrates finite volume method and an embedded discrete fracture model, simulate two-phase flow in 2D or 3D heterogeneous and fractured porous media.	Yan et al. [50]

experiments demonstrate superior performance compared to the B-PINNs [27] and MC-dropout approaches.

He et al. [40] apply PINNs to direct and inverse heat conduction problems of materials. The authors utilize adaptive activation functions and also contribute three improvements to their PINN algorithm, tailored to this class of problems: (i) enhancement of initial values and boundary conditions, (ii) transformation of the time domain in order to handle long-time heat conduction cases, and (iii) a modification to the physical constraint to combat the vanishing gradient problem. In addition, for inverse cases they test two types of neural network schemes with ResNet-inspired skip connections, and investigate inverse problems for conductivity inference for single, multiple, constant, and variable scenarios for use cases of wood and steel. Zhang et al. [141], introduce a multi-domain variant (M-PINN) for steady-state heat conduction in multilayer media, which produces accurate results even for the thin film at nanoscale. It is also shown to outperform the traditional boundary element method for nonlinear heat conduction problems.

Zheng et al. introduce the first order time derivative enhanced parallel hard constraints physics-informed neural network (T-phPINN) [181] for 2D *non-Fourier* heat conduction equations. T-phPINN consists of two sub-networks (FFNs) that train in parallel which predict temperature and its first order time derivative, and is also accompanied by a transfer learning framework and hard constraints. Its design accommodates the governing physical laws that are described by the Cattaneo-Vernotte (CV) model. For their cases, the authors demonstrate that T-phPINN training is significantly faster than FEM; for the first case the computation time of T-phPINN is 33.43% of the FEM computation time, and for the second case it is 51.50%. T-phPINN is also thoroughly investigated against hPINN [16].

Zobeiry and Humfeld [180] investigate heat conduction with convective boundaries, using a carbon-fiber epoxy composite as a representative case. Unlike traditional BCs, their convective BCs enforce PDEs. The proposed enhanced PINN architecture utilizes physics-informed pre-layers for feature engineering, while adaptive activation functions are also investigated. This PINN is compared to vanilla PINNs for 1D and 2D cases, and it showcases enhanced extrapolation capabilities, an observation that highlights the possible advantages of physics-informed architectures and features.

3.1.2 Thermal Convection

Thermal convection is the transfer of heat through a fluid. It can be separated into two categories, forced or natural convection. When both of these mechanisms act together, there is mixed convection. The advection (convection)-diffusion equation describes the transfer of physical qualities into a physical system via diffusion and convection. It is a governing PDE for convection problems. It can be described as:

$$\frac{\partial T}{\partial t} + \mathbf{v} \cdot \nabla T = \alpha \nabla^2 T + S(\mathbf{x}, t) \tag{20}$$

where $\mathbf{v}$ is velocity, α is the diffusion coefficient, and S again represents heat sources/sinks. Note that in the case of $\mathbf{v} = 0$, the advection-diffusion equation becomes the conduction equation Eq. (17). For heat transfer, advection-convection for dimensionless temperature θ can be described as [47]:

$$\frac{\partial \theta}{\partial t} + \mathbf{v} \cdot \nabla \theta = \frac{1}{\mathrm{Pe}} \nabla^2 \theta \tag{21}$$

where Pe is the Péclet number.

In many materials science heat transfer problems involving fluids, the Navier-Stokes equations govern the fluid flow component and are coupled with the governing equations. The incompressible Navier-Stokes equations describe the motion of an incompressible fluid, and are common in the PINNs literature for various problems, including heat transfer-oriented cases:

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla)\mathbf{v} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{v} + \frac{\mathbf{f}}{\rho} \tag{22}$$

$$\nabla \cdot \mathbf{v} = 0 \tag{23}$$

where $\mathbf{v}$ is the velocity vector field, ρ is density, p is pressure, ν is kinematic viscosity and $\mathbf{f}$ represents external forces acting on the fluid. The second equation Eq. (23) is the incompressible continuity equation for mass conservation. There exist several versions of Navier-Stokes depending on the settings and context of the problems at hand.

Cai et al. [47] have provided a seminal article on PINNs and heat transfer problems. In their work, the basics of PINNs applications for heat transfer problems are discussed. The authors examine ill-posed convection scenarios: i) forced convection in an enclosure, ii) forced convection around a circular cylinder in steady-state, and iii) a mixed convection around a circular cylinder. They demonstrate how PINNs can solve these problems by leveraging physics and scarce data, without known boundary conditions.

3.1.3 Phase Change Problems

Phase transition/change problems include cases in which materials change states such as solidification or evaporation. These phase changes cause energy transfer, and they play

an important role in materials science, interacting with the properties and behaviors of the materials.

Phase-field modeling is one of the popular ways to model phase transitions. The following phase-field equations, the Allen-Cahn (AC) equation and the Cahn-Hilliard (CH) equation, are often utilized.

The Allen-Cahn Equation [212] is a reaction-diffusion equation describing the phase separation in multi-component alloy systems:

$$\frac{\partial u}{dt} - \Delta u = -\varepsilon^{-2} f(u), \tag{24}$$

where u is a state variable with a scalar value, $\varepsilon > 0$ and $f(u)$ is commonly $f(u) = u^3 - u$, and it is considered the derivative of a double-well potential $F(u) = 0.25(u^2 - 1)^2$.

The Cahn-Hilliard Equation [213] describes spinodal decomposition:

$$\frac{\partial u}{\partial t} = M\nabla^2 \left(-\gamma\nabla^2 u + f(u)\right), \tag{25}$$

where u now is the order parameter (typically the rescaled density of atoms with values between -1 and 1), M is mobility and γ is related to the surface tension at the interface. $F(u)$ and $f(u)$ are commonly the same as in the AC equation. Both AC and CH are closely related to the *Ginzburg–Landau free energy functional* with AC being the L^2 gradient flow, and CH being the H^{-1} gradient flow. In general, AC and CH are often referenced in the PINNs literature, and they are considered challenging for PINNs. For example, Wight and Zhao [13] discuss the challenges in solving phase-field models with these equations and propose tailored improvements for PINNs (see also 2.3.6), while the evaluation of the bc-PINN variant [185], is also focused on AC and CH.

3.1.3.1 Stefan Problems Stefan problems are challenging free boundary problems in heat transfer, involving a moving interface between two phases (e.g., solid and liquid) that evolves over time, typically driven by a phase change such as melting or freezing. When directly solving Stefan problems, the goal is to track the position of this moving interface, which is not known apriori, and determine the temperature solution. The Stefan problem is typically described by Stefan conditions commonly derived by energy balance/conservation.

The Stefan number (Ste) is a dimensionless parameter that is commonly used when examining Stefan problems. It is the ratio of sensible heat to latent heat, commonly expressed as [188]:

$$Ste = \frac{c_p \delta T}{L}, \tag{26}$$

where c_p is the specific heat, δT is the difference of temperature between phases, and L is the latent heat of melting.

Stefan problem cases are examined in [45], in order to demonstrate the effectiveness of PINNs for such scenarios. Both forward and inverse types of problems are visited, while a data-driven inverse Stefan problem type is introduced. A two-network architecture is applied, where one network estimates the temperature, while the second estimates the moving boundary $s(t)$. Adaptive weights [7] were shown to greatly improve their PINN. Cai et al. [47] utilize this setup as well to demonstrate their Stefan example. Li et al. [187] focus on Stefan problems by also employing a two-network PINN framework, as well as a log-cosh loss function instead of the traditional MSE one. They also demonstrate that small data sampling improves the accuracy of the model after testing several Stefan problem scenarios and synthetic dataset sizes. In [188], the Stefan problem is described by an enthalpy approach and the application of PINNs is examined, while the authors also investigate known PINNs enhancements, specifically pointwise weighting and sequence learning in time.

A PIELM approach specifically for Stefan problems is presented in [153]. The authors present a dual network PIELM framework, while they also employ the iterative least-squares method to address the nonlinearity of Stefan problems. The effectiveness of the method is examined through six case studies, and it is shown that the PIELM approach outperforms the dual PINN framework to a significant degree.

Guo et al. [43] examine the solution of a melting problem with PINNs. Specifically, the authors analyze the melting heat transfer of a non-Newtonian (Sisko) fluid over a moving surface with nonlinear thermal radiation and magnetic fields. For the modeling of the problem, the Buongiorno's model and a nonlinear thermal radiation model are utilized. The authors introduce an adaptive deep collocation method to handle infinity boundary conditions, while they also enhance performance through model-based transfer learning and fitted neural network search.

Rouhollahi et al. [189] examine directional solidification in freeze-casting processes for biomaterial scaffolds. Their model outputs enthalpy, which implicitly encodes both the temperature distribution and solidification patterns. Through numerical experiments, the authors use their PINN to also investigate the effects of (i) scaffold thermophysical properties on solidification and (ii) mold geometry on the solidification rate. The PINN model achieves comparable accuracy to a high-fidelity CFD simulation, with a maximum deviation of 0.8% in the baseline validation case. A

significant advantage of this approach is its computational efficiency: the PINN was trained on only 5,000 data points, a sparse subset equivalent to the information from 250,000 CFD timesteps. This high accuracy, combined with reduced computational cost and less dependency on predefined boundary conditions, makes the proposed approach advantageous for real-world applications.

Although heat transfer dominates many phase-change processes, transitions can also occur due to mechanical or chemical effects rather than thermal influence.

3.1.4 Radiative Heat Transfer

Thermal radiation is also a fundamental mechanism of heat transfer. It is emitted by the thermal motion of (matter) particles. We recommend Modest's book [214] for an in-depth analysis. The following formulas for the radiative transfer equation (RTE) are derived from [214].

The RTE describes how the spectral radiative intensity changes:

$$\frac{dI_\eta}{ds} = \kappa_\eta I_{b\eta} - \kappa_\eta I_\eta - \sigma_{s\eta} I_\eta + \frac{\sigma_{s\eta}}{4\pi} \int_{4\pi} I_\eta\left(\hat{\mathrm{s}}_i\right) \Phi_\eta\left(\hat{\mathrm{s}}_i, \hat{\mathrm{s}}\right) d\omega_i \tag{27}$$

where s is a path, $\hat{\mathrm{s}}$ is direction, ω is solid angle, η is wavenumber and the intensity is augmented by emission ($\kappa_\eta I_{b\eta}$), and decreased by absorption ($\kappa_\eta I_\eta$) and by scattering away from $\hat{\mathrm{s}}(\sigma_{s\eta} I_\eta)$. Φ is the scattering phase function.

Transient RTE:

$$\frac{1}{c}\frac{\partial I_\eta}{\partial t} + \frac{\partial I_\eta}{\partial s} = j_\eta - \kappa_\eta I_\eta - \sigma_{s\eta} I_\eta + \frac{\sigma_{s\eta}}{4\pi} \int_{4\pi} I_\eta\left(\hat{\mathrm{s}}_i\right) \Phi_\eta\left(\hat{\mathrm{s}}_i, \hat{\mathrm{s}}\right) d\omega_i \tag{28}$$

where c is the speed of light and j_η represents an emission (source) term.

Mishra and Molinaro [190] investigate the application of PINNs for radiative transfer problems, employing the RTE as the governing equation. They demonstrate PINNs' capability in solving both forward and inverse radiative transfer problems, validating their approach through benchmark cases, including monochromatic and polychromatic radiative transfer in 1D and 3D, as well as time-dependent simulations of radiative transport from a radiating sphere into a cold, surrounding medium.

On the other hand, Riganti and Dal Negro [46] address integro-differential radiative heat transfer problems, including inverse and coupled cases, by employing auxiliary PINNs (A-PINN) [186]. They efficiently solve the Boltzmann-type transport equations using slab geometry and scattering media described by various types of phase functions. In addition, they demonstrate the application of A-PINN to the coupled radiation-conduction problem in participating media, revealing temperature profiles beyond the Fourier thermal conduction limit. The paper also tackles the Schwarzschild–Milne integral equation (that describes isotropic scattering processes) in an inverse setting, retrieving the single scattering albedo using only boundary data.

Murari and Sundar [44] examine the PINNs approach for radiative heat transfer in a graded-index medium. The RTE with graded index is particularly challenging with its high-dimensional nature, resulting in challenges or failures when using traditional approximation methods. The authors show that PINNs can be a healthier alternative.

3.1.5 Multiphysics and Realistic Applications

Cai et al. [47] demonstrate how PINNs can be applicable for heat transfer analysis in power electronics with two separate cases. First, the ANSYS group provides several experimental results for heat transfer in electronic chips, chip tile-based temperature prediction, as well as solutions generated by their hybridized Auto Encoder and Image Gradient based network [215] for 2D and 3D chip thermal analysis. For the second case, heat sink designs for parameterized geometries are provided by NVIDIA. These are produced by the toolkit 'SimNet' [216], a PINN-based framework. Such examples underline the role PINNs could play at the industrial level of applications.

Chandan et al. investigate radiative heat transfer in a concave porous fin under local thermal non-equilibrium (LTNE) conditions [191]. They employ the LTNE model to describe the separate energy equations for the solid and fluid phases in the porous medium. Conduction, convection, and radiation are incorporated into the heat transfer framework. The governing equations are solved using two approaches: PINNs and the Clique Polynomial Method (CPM). Their results demonstrate that PINNs effectively solve the problem, as both methods yield consistent and satisfactory solutions.

Jalili et al. [51] investigate two-phase flows with heat transfer. Their main focus is the case of the rising bubble with a hot wall, a multiphysics problem involving several interactive phenomena. The authors investigate several different scenarios in order to gain insights for the interacting phenomena, and then approach the rising bubble case with a tailored, multi-network PINN framework. The framework enforces coupled PDEs (mass conservation, momentum transport, and energy conservation), data-based losses, and physical constraints and is assessed for its forward, inverse, and extrapolative prediction capability, while the computational cost efficiency (compared to CFD) is highlighted.

In [192] PINNs are applied to model the thermochemical evolution of a composite material on a tool undergoing cure in an autoclave. For this approach, the governing coupled equations: conductive heat transfer and resin cure kinetics are

handled by a dual, disconnected architecture, while the training method is sequential in order to mitigate instability. The PINN is examined for single and bi-material settings. The framework is enhanced by a modified adaptive loss weights strategy, transfer learning, and additional inputs for surrogate modeling, and the effects are thoroughly investigated.

Kim et al. [52] utilize a multiphysics PINN (MPINN) approach for the modeling and prediction of thermal runaway (TR) in lithium-ion batteries (LIBs), a critical safety issue. The MPINN models TR, with coupled heat transfer and heat generation from exothermic reactions. Two key reaction models are investigated separately, governed by the Arrhenius law; the positive electrode decomposition model and the electrolyte decomposition model. This MPINN architecture consists of two separate NNs, estimating temperature and dimensionless concentration of reactive species respectively. The authors examine the MPINN's capabilities regarding semi-supervised learning, unsupervised learning, and surrogate modeling, while the novel MPINN is shown to outperform data-driven NNs.

Liu et al. [193] use PINNs for multiscale modeling, specifically for a Polyurethane -Phase Change Materials (PU-PCM) foam composite, a material with advantageous behavior for building insulation and indoor thermal stability. The authors employ PINNs in their hierarchical multiscale model for thermal conductivity prediction, and then extend their strategy to optimize the design process. Their framework employs PINNs for micro-scale modeling, an RVE-FEM (Representative Volume Element - Finite Element Method) approach for meso-scale modeling, and the Mori-Tanaka method for macro-scale modeling. To evaluate the optimized PU-PCM material, they present a case study of a detached house, showcasing good thermal insulation and improved energy consumption.

Another prevalent multiphysics problem is thermo-hydro-mechanical (THM) processes. Amini et al. investigate THM coupling in porous media using PINNs [194]. This challenging setting involves several coupled PDEs that govern balance laws, deformation kinematics, poro-elasticity, two-phase flow, and heat transfer. The authors employ non-dimensional forms of the equations and apply enhancing strategies, notably sequential training, to tackle the optimization challenges. They successfully simulate four distinct benchmark cases (including consolidation and injection-production scenarios), achieving good agreement with reference solutions, while also emphasizing the inherent difficulties in training PINNs for such strongly coupled, multiscale problems.

3.1.5.1 Additive Manufacturing Zhu et al. [53] introduce the application of PINNs to a multiphysics process, metal Additive Manufacturing (AM). Metal AM inherently involves several physical interactions between different phases, resulting in a complicated, multiscale, and multiphysics phenomenon. The authors examine their PINN framework for two manufacturing problems: i) the solidification process of aluminum in a graphite mold, and (ii) the first challenge of the test series included in the 2018 NIST Additive Manufacturing Benchmark (AM-Bench). The problem corresponding to this challenge is a selective laser beam melting process of a pure Inconel 625 (IN625) substrate. The AM processes are in three dimensions. The goal is to predict the temperature, melt pool fluid dynamics, melt pool dimensions, and cooling rates, while three separate laser configurations are examined. The PINN framework is evaluated and produces good approximations. Another AM-related investigation is performed by Xie et al. [54], regarding the direct energy deposition process (DED). The authors employ PINNs for 3D temperature field prediction during DED, which is considered vital information. The PINN approach is evaluated both for single-layer and multi-layer DED, while the FEM method is used to generate training and testing data. For the experimental validation of the PINN model, the authors conducted DED experiments of Inconel 718, and an infrared bicolor thermometer was used for real-time measurements. For three different settings, the mean relative error for temperature prediction was 4.83%, 2.85% and 3.21%. The authors also conduct investigations for 316 L stainless steel to check the generalization of their strategy, and they also showcase that their PINN framework can offer the same precision as a data-driven network or XGBoost while using 20% of the required training data.

A different approach regarding PINNs for metal AM temperature prediction, is presented by Sajadi et al. [195]. Their framework employs physics-informed online learning, where a PINN is first pre-trained on baseline data and then dynamically updates its weights during an online phase using new process data. This allows the model to adapt to unseen scenarios in real-time, enhancing flexibility. The study demonstrates superior performance compared to a traditional data-driven network, particularly in accurately predicting temperatures in critical regions like the Heat Affected Zone (HAZ) and melt pool.

Another AM technique is Stereolithography (SLA), which relies on the exothermal photo-polymerization process. In [196], Tod et al. model the thermal behavior of a SLA process during the cooling phase, by following a unique PINN-based strategy. A data-driven NN is trained with temperature data, so that it can provide temperature measurements and derivatives to a second NN that is used for physics discovery. Then the second NN is used as a regularizer for the PINN during training. This PINN provides physical insights for the construction of an improved

physical model. The improved model and the PINN are shown to outperform the initial model (linear heat equation) for two showcased geometries. The acquired physical model can produce results in 10s, a significant advantage highlighted by the authors. This unique strategy showcases that PINNs can also be used for physical model design.

3.2 Materials Properties & Design

Materials properties and design, including component or device design, are the core aspects of materials science and modeling. Several examples exist in the PINN literature, representing different use cases or physical phenomena. The following examples are divided according to each the *main* focus and application of each paper, but it is important to note that in many cases property inference, material behavior, and design go hand in hand, or even accompany further analysis. Properties can be related to several physical aspects of the material, such as thermal, mechanic, electromagnetic, and optical - photonic properties.

3.2.1 Materials Properties

3.2.1.1 Mechanical Properties Rezaei et al. [197] introduce the COMM-PINN (Constitutive Material Modeling with Physics-Informed Neural Networks) framework. This approach enables the solution of constitutive laws, encompassing yield conditions, evolution laws, and Karush–Kuhn–Tucker (KKT) conditions directly from physical principles without requiring labeled data. The resulting trained network functions as a user-defined, thermodynamically consistent material model, predictive for arbitrary loading paths within the specified collocation domain, and suitable for integration into finite element codes or other numerical simulators. Although focused on material modeling, this framework leverages constitutive laws fundamental to solid mechanics [217].

Lee and Popovics's PINN framework [32] solves the linear wave equation and is utilized for the inference of in-place properties of complex materials such as mortar. The authors use mechanical wave propagation data, specifically a mapping of the displacement acquired by an air-coupled transducer (ACT), to acquire a training set, while their implementation also utilizes sub-networks to estimate coefficients of the embedded wave equation. Although this scenario is investigated in one dimension, the authors note that it can follow the same principles for added dimensions.

A more unique case, in the broader scope of constitutive laws, is a non-Newtonian material property, thixotropy. Zhang et al. introduce the PINN-based RheologyNet framework [33] for the evaluation of thixotropic properties of cementitious materials. RheologyNet relies on rheological PDEs to predict the shear stress, the apparent viscosity of the thixotropic liquid, and the internal structure parameter. The authors propose two quantitative indexes derived from the predicted rheological behavior: thixotropy area and thixotropy variability. Through several investigations evaluated against Finite Element Analysis (FEA), RheologyNet is shown to significantly outperform data-driven FNNs and demonstrate accuracy and robustness. In the final set of numerical experiments, the mean relative error for thixotropy area estimation across different water/cement ratios is 5.8×10^{-4} for RheologyNet, while the data-driven NN achieves 4.2×10^{-3}.

3.2.1.2 Elastic Properties Haghighat et al. [34] provide one of the earliest applications of PINNs for elasticity. They investigate PINNs for linear elasticity cases, and extend their investigation to an example with von Mises elastoplasticity. For their approach, multi-network PINNs are shown to better capture the field variables, while transfer learning and surrogate modeling are also investigated.

In [198], Vahab et al. employ PINNs for solving biharmonic problems of elasticity, namely the Lamé problem, the semi-infinite foundation problem, and elastic plate theory problem cases. The authors propose distinct strategies inspired by the classical analytical solutions of the Biharmonic equation that demonstrate improved performance. Instead of directly approximating stress/displacement, the Airy function ϕ is used, while the 4th-order biharmonic equation is reduced to two 2nd-order equations, tackled by two conjugate networks. Fourier series are used to represent the approximated solution for a specific case. Thorough investigations of these approaches are provided.

A different case is presented in [31], where PINNs are used for the identification of nonhomogeneous materials for the biomedical scenario of the elasticity imaging problem. This approach allows for the distinction between normal and diseased tissues. A two-network PINN framework is proposed, and the method is validated for the prediction of the shear modulus field with a prototypical plane strain problem for incompressible, hyperelastic tissue. Anton and Wessels [199] employ PINNs for the calibration of linear-elastic constitutive models from full-field displacement and global force data in a realistic setting. Their PINN is refined via normalization, scaling, global information and by formulating the constitutive model in terms of bulk and shear modulus instead of directly estimating Young's Modulus and Poisson's Ratio, an approach that showed improved results.

3.2.1.3 Optical and Photonic Properties Chen et al. [35] apply PINNs to solve representative inverse scattering problems in photonic metamaterials and nano-optics technologies. The authors utilize PINNs to obtain optical properties, specifically electric permittivity parameters, with respect to various finite-size scattering systems. They showcase how the PINN framework can reconstruct the spatial distribution of the electric permittivity of unknown objects from synthetic scattering data, while they also apply PINNs for a cloaking problem case. Chen and Dal Negro expand their investigations in [36] to apply PINNs for problems of near-field microscopy. Their PINN framework integrates the full-vector Maxwell's equations to solve inverse problems; specifically to obtain the spatial distributions of the complex electric permittivity and magnetic permeability of nano-structured environments. An adaptive loss weight scheme is proposed, while the efficiency and accuracy of the proposed PINN implementation are examined. The authors verify its capabilities by successfully retrieving the space-dependent permittivity of a three-dimensional unknown object from near-field data.

PINNs solve the inverse problem of the nonlinear Schrödinger equation (NLSE), to infer optical fiber parameters (attenuation, dispersion, and nonlinear coefficient) in [200]. The available labeled data consist only of the transmitted and received waveforms. This PINN predicts the real and imaginary parts of the complex envelope of a slowly varying optical field, while the launch power is also embedded as a parametric feature to allow for more efficient training, richer data representation, and enhanced generalization ability.

Wang et al. [201], present the first application of PINNs for fiber amplifiers. The governing equations for the prediction of pulse amplification with PINNs are the dynamic rate equations (DREs) and power transmission equations (PTEs), with outputs including the forward/backward optical powers and the upper level population concentration. The authors demonstrate their PINNs for inverse problems for the retrieval of system parameters, such as pump/signal overlap factors and pump power, and highlight the usefulness of such PINNs as components of system detection and monitoring frameworks.

3.2.1.4 Electromagnetic Properties Khan et al. [48] present a feasibility study of PINNs for electromagnetic problems. They investigate PINNs applications for three forward problems of such nature, (i) a simple 2D electrostatic box, (ii) its extension to two materials sharing an interface, and (iii) a multi-domain magnetostatic case of a copper coil located in a cylinder filled with ferromagnetic material. The study analyzes solution accuracy, the impact of transfer learning to accelerate training, and the benefits of hybrid training with labeled data. Hu et al. [37] propose an unsupervised PINN framework for solving electromagnetic inverse scattering problems, aiming to reconstruct complex permittivity distributions from field data. The authors propose a frequency scale factor to avoid gradient explosion due to the multifrequency data, and also enhance their PINN with various strategies. This PINN is directly compared to Chen's approach in [35], and is shown to outperform it, while providing comparable or improved results to a traditional inversion method (FIM), showing greater generalization capability, especially for cases of high-contrast and electrically large scatterers.

Kovacs et al. [38], investigate magnetostatic and micromagnetic problems. They use standard collocation-based PINNs and the Deep Ritz Method [97] which also belongs to physics-informed ML. They present three test cases, with the PINNs application being applied to two of them: (i) computing the magnetic field of a uniformly magnetized infinite prism, and (ii) an inverse problem, the estimation of the magnetization distribution for a Halbach cylinder. The authors highlight these approaches as versatile, efficient, and suitable for these types of problems.

For a realistic problem case, Son et al. use a problem-tailored PINN framework to model the electromagnetic response of a permanent magnet synchronous motor [202]. Their novel architecture accounts for three partial differential equations with rotational coordinate transformation, decomposition to separate the rotor and stator domains, and a tailored interface loss term, while it also utilizes adaptive weights (learning rate annealing) [7]. Their framework is evaluated against both finite element analysis and experimental measurements, while both normal and failure conditions were considered. For the five operational cases, the PINN framework's inference time is shown to be at least 10 times faster than the finite element analysis. When compared against experimental data, the average root mean square error (RMSE) in torque estimation was 0.0105 Nm for FEA and a comparable 0.0166 Nm for the PINN, highlighting the latter as a promising and highly efficient tool.

3.2.2 Materials Design

3.2.2.1 Optical and Photonic Materials Riganti et al. [58] employ multiscale PINNs (MscalePINNs) for the inverse design of finite-size photonic materials, specifically targeting arrays of dielectric nanocylinders with stealthy hyperuniform (SHU) and uncorrelated Poisson disordered geometries. Their method infers the spatially dependent permittivity profiles of SHU and Poisson structures. Notably, the authors demonstrate that SHU arrays can be homoge-

nized effectively and isotropically at the single-realization level, bypassing the need for disorder averaging. This capability facilitates the design of angle-insensitive photonic media, waveguides and binary metamaterials. The authors also highlight MscalePINNs' superior performance over conventional PINNs for cases where significant multiple scattering effects determine the effective parameters.

3.2.2.2 Electromagnetic Materials Fang and Zhan [203] propose PINNs for electromagnetic metamaterials design. Metamaterials demonstrate unconventional, desirable electromagnetic traits and can be used for many types of practical applications. Their PINN inversely solves the frequency domain Maxwell's equation to design permittivity and permeability. The framework is applied for a cloaking metamaterial problem, which is significantly improved by a piecewise constancy approach, and a rotator design case. In this work, the PINN produced novel solutions for metamaterial design for cloaking, and they were proven to be reliable by validation with the Comsol solver.

Baldan et al. [59], introduce a novel methodological approach for PINNs regarding inverse electromagnetic problems. The focus of this paper is magnetostatic problems. The authors introduce a hypernetwork (HNN) which produces the PINNs parameters (weights) for the corresponding system parameters. The HNN training process is complex and demanding, but does not require repetition. The produced PINNs immediately solve the direct problem. After training, this HNN-PINN framework acts as a real-time field solver, allowing for fast design with optimization algorithms. This strategy is applied to a coil optimal design task.

3.2.3 Multiscale Materials Modeling

The analysis and design of multiscale materials and metamaterials is a significant challenge, for which the modeling of cross-scale information interactions is vital. PINNs often fail to properly capture multiscale behaviors and high frequencies, with spectral bias being a major contributor to this drawback. Thus, several novel PINN applications focus on this specific challenge.

Wu et al. [60] propose a PINN-enhanced Extended Multiscale Finite Element Method (EMsFEM). In EMsFEM the domain is discretized into two sets of meshes (coarse-scale and fine-scale) to obtain the multiscale solution and predict microscale behaviors, respectively. One major computational drawback of this method is the construction of numerical base functions that bridge the two scales by describing the material's heterogeneous properties. The authors employ PINNs with domain decomposition to map fine-scale elements to numerical base functions. This approach replaces the need to solve numerous expensive boundary value problems with a rapid neural network inference. As a result, the proposed framework is more computationally efficient than conventional EMsFEM. Specifically, for the cantilever beam study case, the computational time of EMsFEM is 244.68s, while for EMsFEM-PINN, it is 22.13s. In addition, the results produced by the two approaches are consistent with each other over all the geometries tested, with relative errors between them being less than 1.00%. EMsFEM-PINN also exhibits strong generalization since it does not require retraining for different global geometries or loading conditions.

The Neural Homogenization-based PINN (NH-PINN) [61] is also a multiscale-oriented approach. Homogenization allows the approximation of multiscale equations by homogenized equations without multiscale characteristics. The homogenized equations can capture the macroscopic behavior of the solution. The framework uses PINNs to execute the three-step homogenization process that translates to: (i) solving the periodic cell problems, (ii) evaluating the homogenized coefficients (directly through AD), and (iii) solving the final homogenized PDE. An oversampling method is proposed for the accurate solution of the periodic cell problems. Several scenarios employing PINNs and FEM are investigated and used as benchmarks to examine and validate NH-PINN, and according to their observations, the authors highlight that PINNs are a promising new approach for homogenization instead of numerical-driven methods.

Jeong et al. [204] propose the Complete PINN-based Topology Optimization (CPINNTO) framework, a specialized approach for complex TO problems, including multiscale, multi-material, and geometrically nonlinear design. CPINNTO replaces the standard structural (FEA) and sensitivity analyses relative to TO, with two PINN variants: Deep Energy Method PINN (DEM-PINN) and Sensitivity-analysis PINN (SEN-PINN)[1]. The SEN-PINN generates density fields of the problem domain with its loss function satisfying minimized compliance and target volume fraction, while the DEM-PINN predicts corresponding displacement fields by minimizing the overall potential energy of the system. The outputs of these networks participate in each other's losses. When using CPINNTO for multiscale TO, the authors propose the concurrent design of macrostructure and its composite microstructure. The recommended strategy employs two SEN-PINNs, dedicated to macrostructure and microstructure respectively, while the energy-based

[1] In the original paper, the Sensitivity-analysis PINN is referred to as S-PINN. Due to the symmetry-preserving PINNs also being abbreviated as S-PINNs, an alternative is used here.

homogenization method determines the effective elastic properties of microstructure. CPINNTO and its applications are validated for several complex benchmarks, highlighting the feasibility of PINNs for TO. However, significantly higher computational times, as well as some undesired behaviors are observed. To this end, future evolutions of the framework may suggest improvements.

3.2.4 Materials Durability

Materials durability is a vital aspect of materials modeling. In order to assess the related characteristics and behaviors of materials, several approaches can be adopted. In the following paragraphs, the referenced literature is organized into three thematic groups; defect analysis, fracture and crack propagation analysis, and fatigue life.

3.2.4.1 Defect Analysis Shukla et al. [55], employ PINNs as a nondestructive quantification method and characterize surface breaking cracks in metal plates. Based on the premise that wave speed variations indicate crack presence, their framework uses ultrasonic wavefield data and solves the inverse acoustic wave equation via PINNs to determine the spatially varying wave speed, thereby locating the crack through regions of reduced speed. The study also assesses the minimum experimental data required for accurate identification.

In a different approach, Zhang et al. [205] develop a PINN-based inverse framework to characterize internal structures and defects. The authors apply their approach on rectangular matrix materials with unknown voids - inclusions. Using only boundary displacement data, the method identifies unknown geometric parameters (size, shape, location) of the voids - inclusions as well as the shear modulus of the inclusion for one of the examples. The framework is validated against FEM.

Lim et al. [56] use PINNs as a nondestructive testing method for carbon fiber-reinforced polymer (CFRP). Their framework models Fourier's law of heat diffusion to process pulsed thermography data. These PINNs estimate nonuniform background through a grayscale value that represents temperature. Subtracting this prediction from the original thermograms yields residual images where defects are enhanced. Afterward, a principal component thermograph (PCT) method is applied for dimensionality reduction, allowing efficient defect characterization. The proposed framework is also tested for inverse settings, estimating thermal diffusivity, while it is evaluated against a data-driven neural network, surpassing its predictive capacity for this case.

3.2.4.2 Fracture and Crack Propagation Analysis Goswami et al. [57] introduced the first PINN application for the phase field modeling of fracture, especially for brittle fracture problems. The prediction of the crack path is a challenging problem that often causes traditional methods to struggle. This specialized PINN minimizes variational energy and integrates hard BC constraints and Gauss quadrature rules. Simulating fracture propagation is done incrementally; therefore, retraining is needed for each displacement step. The authors recommend a transfer learning approach that significantly reduces computational costs. This tailored framework yields results that have been previously verified, while the variational energy approach is shown to outperform vanilla PINNs.

Tu et al. [206] investigate crack propagation for industrial applications. They also minimize variational energy. The authors propose a PointNet-based adaptive refinement method to bypass the need for precalculation when constructing the discrete domain and thus efficiently obtain Gaussian integration points. This approach is promising for future computationally light PINN applications in crack propagation modeling. In [207], the Enriched PINN (E-PINN), which is tailored for 2D in-plane crack analysis, is introduced. The E-PINN incorporates crack-tip enrichment functions in order to precisely model the local behavior in the crack-tip region. The approach is investigated for a number of benchmarks for loading and crack-mode types and showcases good approximations of stress intensity factors (SIFs) with only a few degrees of freedom.

3.2.4.3 Fatigue Life Fatigue life analysis and estimation is an area strongly correlated with the study of defects and/or fractures in materials. Dong et al. [208] apply PINNs to predict the lifetime of multi-defect materials under cyclic loading. The authors leverage fatigue data from the literature in order to effectively train their PINNs. They propose two fracture-based PINN approaches, namely FP-M, and FP-P, both leveraging concepts from the M-integral fatigue for their physical loss. For the FP-M model, the physical loss function integrates the model itself, while for the FP-P model, the physical loss function embeds three expressions of physical knowledge derived from the model. The two models are evaluated against data-driven networks for No.45 Steel and Al-Mg4.5Mn alloy and both showcase a squared correlation coefficient of over 0.94, while the best value the data-driven NN reached was 0.5451 for the steel case. Fatigue life is also investigated by Salvati et al. [209], specifically for additively manufactured metallic materials with defects; a factor causing uncertainty in estimations. The authors propose a novel, semi-empirical Linear Elastic Fracture Mechanics (LEFM) model for the physical

constraints, which combined with the PINN and properly obtained data constitute the fatigue estimation framework. A study case for an AlSi10Mg alloy is used to evaluate the strategy, and the PINN is validated by a K-fold cross validation method.

3.2.5 Stress Analysis

Stress analysis employs engineering techniques to determine internal stresses arising from external loads, as well as from more complex multiphysics sources such as the diffusion of species within a material.

In [62], multi-output PINNs are applied to solve boundary value problems in continuum micromechanics with inhomogeneous material distributions. The authors compare three approaches: (i) standard PINN, (ii) a domain decomposition approach (cPINN [26]), and (iii) PINN and cPINN enhanced by an adaptive discretization scheme, all designed to resolve displacement, stress, and energy fields in arbitrary microstructural unit cells. The authors evaluate these approaches through test cases, for which the cPINN approach using regular sampling is the most successful. This approach is further investigated and tested for a real-world μCT-scan of a wood-plastic composite, producing accurate solutions.

Xue et al. [63] apply PINNs to solve diffusion-induced stress in an elastic sphere, a problem relevant to lithium-ion battery degradation. The governing equations describe mass transport and mechanical equilibrium, and the problem is formulated with or without hydrostatic stress-limited diffusion, the first approach being the more physically complete scenario.

3.3 Porous Media and Multi-Phase Materials

Fluid flow in porous media is a standard way to characterize porous media, a huge class of materials with various applications and processes relevant to materials science. In the literature, several PINNs applications are presented for such cases, which are often multiphysics-based in nature. The following are such cases.

Darcy's Law is often adopted to model flow in porous media. The fundamental Darcy Law expression is as follows:

$$\mathbf{q} = -\frac{k}{\mu}\nabla p \tag{29}$$

where $\mathbf{q}$ is the volumetric flux , k is a constant describing the permeability of the medium, μ is the dynamic viscosity of the fluid, and the term ∇p is the pressure drop.

Here we also provide the steady-state formulation as seen in [154] for average pore velocity.

$$\mathbf{v}(\mathbf{x}) = -\frac{K(\mathbf{x})}{\phi}\nabla h(\mathbf{x}) \tag{30}$$

$$\nabla \cdot \mathbf{v}(\mathbf{x}) = 0, \quad \mathbf{x} \in \Omega \tag{31}$$

in this derived equation, ϕ is the effective porosity of the medium, $\mathbf{v}$ is the average pore velocity for which $\mathbf{v} = \frac{\mathbf{q}}{\phi}$, $K(x)$ is conductivity, and $h(\mathbf{x})$ is hydraulic head. The second equation enforces continuity.

Tartakovsky et al. [210] apply PINNs to subsurface flow problems. The PINNs are constrained by Darcy's Law to estimate hydraulic conductivity, or Richard's Equation for the case of unsaturated flow. Their two-PINN approach outperforms purely data-driven networks. The authors demonstrate that for the case of unsaturated flow in homogeneous porous media, PINNs successfully model the relationship between pressure and conductivity, with the provided data being only capillary pressure measurements, even in the presence of noise. He et al. extend this work to include transport phenomena described by advection–dispersion equations, thus the case of subsurface flow and transport [154]. The authors introduce a three-network MPINN (multiphysics PINN) that jointly trains for the space-dependent hydraulic conductivity, hydraulic head, and concentration fields, respectively. It is compared with the PINN-Darcy approach in [210], as well as with three independent data-driven NNs (one for each physical quantity). The authors investigate these approaches, showcasing that MPINN outperforms its competitors, and propose that by inverting the parameters of interest jointly, greater accuracy can be achieved for multiphysics settings.

Almajid and Abu-Al-Saud apply PINNs to a classic, fundamental problem in porous media flow: the Buckley-Leverett problem for two-phase displacement [211]. This work examines the one-dimensional case. The authors systematically examine several scenarios (varying data availability, parameter trainability, and inclusion of artificial diffusion), across both small and large end-point mobility ratios. The study includes inverse problem investigations (inferring flow parameters) and explicitly compares PINN performance against standard neural networks, demonstrating the advantage of physics-informed models, particularly regarding data dependency and extrapolation capabilities.

Chen et al. [49] employ PINNs to model resin flow in fibre textiles, a problem relevant to composite manufacturing. For this study, the authors examine PINNs for more realistic and demanding scenarios. Their PINN is governed by Newtonian flow and Darcy's Law for

two-phase flow, which are also non-dimensionalized. The multi-network PINN estimates four variables: pressure, saturation, and the two velocity components. The PINN is evaluated for its ability to handle realistic physical parameters and heterogeneous permeability fields in a 2D linear injection configuration. Three distinct scenarios are tested: (i) sinusoidal permeability profile, (ii) race tracking, and (iii) circular defect, while the PINN is evaluated against a Finite Element/Control Volume solver, revealing promising potential for future real-world applications.

Zhang et al. investigate fluid flow in porous media by employing PICNNs, which are physics-informed convolutional neural networks [124, 125]. The networks' inputs are grid-based image-like data; specifically 2D reservoir models. Initially [124], transient single-phase Darcy flows are examined for highly heterogeneous reservoir models with source/sink terms, and this case is extended for two-phase Darcy flows in [125]. The Finite Volume Method (FVM) is integrated in the PICNN for the PDE residual calculation. In addition, a well model is included in the loss function to represent the high pressure gradient near source/sink terms. The PICNN builds pressure field solutions step by step. In the 2022 paper, CNNs are stacked to form a deep PICNN, while in the 2023 paper, the same CNN produces solutions sequentially. Both these approaches enforce transfer learning. The authors also investigate the use of PICNNs/ Deep PICNNs as surrogates, and demonstrate results for both homogeneous and heterogeneous cases. Yan et al. [50] also investigate reservoirs, aiming to surpass the PICNN approach. They introduce an enriched PINN, namely E-PINN, for the solution of two-phase flow in 2D or 3D heterogeneous and fractured porous media. E-PINN shares some similarities with PICNN, since it also utilizes FVM, but is enhanced with several strategies to further optimize it for this class of problems. These include the Embedded Discrete Fracture Model (EDFM) to represent fractures and a specialized FNN architecture with adjacency-location anchoring, adaptive activation, skip connections, and gated updating. E-PINNs are shown to outperform PICNN. For example, for a 3D heterogeneous reservoir, the simulation time was reduced from 17291s (PICNN) to 2796s (E-PINN). More critically, regarding accuracy, it is shown that for the same instance of simulated time E-PINN maintained a relative error of less than 1% for both pressure and saturation fields, while PICNN's saturation error exceeded 20%. The authors also demonstrate how E-PINNs simulate immiscible displacements in 2D/3D fractured reservoirs demonstrating adaptability to complex practical problems.

4 Discussion

4.1 Advantages

PINNs constitute a physics-informed deep learning method that offers leverage in relation to data dependency. PINNs reduce the reliance on large volumes of labeled data by incorporating physical constraints, which is crucial for cases where data acquisition is expensive or unattainable. Thus, they offer a rather elegant, hybrid approach to learning, which potentially offers better generalization capabilities. One of the most important advantages of PINNs is their suitability and performance for inverse or ill-posed problems. Traditional methods often struggle or fail at such tasks, despite their superior performance in well-posed forward problems. Furthermore, standard MLP- based PINNs are also mesh-free [4, 5], which can offer significant advantages over traditional mesh-based methods like FEM. This mesh-free nature is particularly powerful for high-dimensional problems, where the curse of dimensionality makes mesh generation unrealizable, and for problems with geometrically complex domains, where mesh generation can be a major engineering bottleneck.

It is important to consider PINNs as tools that coexist with traditional numerical methods. They are not necessarily antagonizing established approaches, but rather highlight that these different methodologies offer distinct advantages and drawbacks and are valuable in the scientific community. Additionally, PINNs can integrate data produced by traditional methods in the data-driven part of their learning, or even be combined with them.

As mentioned in previous chapters, a properly defined PINN also offers physical interpretability, unlike data-driven NNs, whose mapping is considered more of a black box. This facilitates the investigation of complex physical phenomena. Additionally, since PINNs are not fully dependent on data, the physical constraints may offer inherent regularization against noise, although careful tuning might still be required.

Furthermore, there is a growing arsenal of strategies that can be used to enhance PINNs and their training. In combination with the flexibility present in PINN configurations and the growing collection of PINNs extensions and variants already serving specific purposes, PINNs can be tailored for specific needs on a theoretical or industrial level.

According to the trends presented in the literature and the fast growth of PINNs, it is expected that these frameworks will keep evolving rapidly, while new methods or variants might offer insights and better solutions for classes of problems. The diversity of PINNs presented in this study is meant to encourage researchers on their investigation and application of PINNs.

4.2 Challenges

Although PINNs have been providing promising results, they also often fail to solve PDEs, especially in multiscale, chaotic or complex settings [5, 110, 116]. In addition, theoretical work regarding their convergence, behavior and error bounds does exist, for example, for linear PDEs [218], but nonlinear cases remain a challenge with no guaranteed convergence [219].

It has also been mentioned (see Section 2.3.1) that PINNs present pathologies that may cause failures. Several studies have investigated PINNs pathologies and failure modes, concerning: unbalanced back-propagated gradients [7, 8], the ill-conditioned loss landscape [108, 163], spectral bias [8, 123, 220], and violation of temporal causality [116]. Cauchy problems also raise challenges for PINNs and NNs in general [221]. However, *to a degree*, many of these pathologies can be countered by the integration of suitable strategies and configurations such as those presented in Section 2.3. It is important to note that they cannot be completely eradicated.

Another important point of discussion is how the mesh-free element can be considered a double-edged sword. The mesh-free characteristic has historically been presented as a strong advantage of PINNs, allowing for flexibility and less computational load. However, typical mesh-free MLPs lack any inherent inductive bias for modeling spatial relationships. This limitation means that a PINN without an architecture designed for spatial reasoning (e.g., convolutional operators) cannot guarantee the inference of neighboring relations and might not be able to properly capture the underlying physics resulting in poor performance. For several types of materials modeling problems, for which structural and relational information is often vital, this shortcoming can be particularly impactful, and appropriate strategies should be adopted.

Furthermore, complex, multiphysics cases might need careful considerations regarding the choice of architecture in order to capture the features and underlying physics of intricate phenomena with good accuracy. Some of the examples presented in this paper (such as [51, 154, 192]), demonstrate that a tailored network configuration is preferred. For inverse problems, if the parameters of interest are not constant, they introduce extra complexity to the problem, which also needs to be addressed. Thus, as it often is with neural networks, the PINN configuration and architecture can be considered a design task in itself and a trade-off between flexibility-accuracy and resources allocated for design investigation.

Generally speaking, PINNs remain NNs and are not immune to challenges such as overfitting and high computational or hardware demands. Optimally training a neural network is a multifaceted challenge, and training time is a significant drawback.

Beyond technical pathologies, a broader challenge is the ongoing debate regarding methodological rigor in ML publications. For example, a recent critical review by McGreivy and Hakim [222], discusses flawed comparisons between the performances of ML-based PDE solvers and traditional methods present in the scientific literature. Through the examination of ML-based methods for forward PDE cases, the authors identify two major pitfalls leading to overoptimism: (i) comparisons with weak baselines, and (ii) various reporting biases that often manifest as the omission or misrepresentation of computational costs. Although this critique does not examine PINNs in the inverse, high-dimensional or geometrically complex contexts where their advantages are often more pronounced, it methodically highlights these pitfalls and their patterns both for ML methods and PINNs. Such observations raise serious concerns and emphasize the need for greater transparency in the scientific community. To this end, systematic adoption of stronger baselines, contextualized computational costs, and fair benchmarking studies is recommended for all ML/PINN frameworks to ensure balanced assessment of their performance.

Table 4 Coding libraries and tools for PiML/PINNs

Software Library	Website - Repository
PyTorch [223]	https://pytorch.org/
TensorFlow [224]	https://www.tensorflow.org/
JAX [225]	https://github.com/jax-ml/jax
PaddlePaddle [226]	https://www.paddlepaddle.org.cn/en
DeepXDE [11]	https://github.com/lululxvi/deepxde
SciAnn [227]	https://github.com/ehsanhaghighat/sciann
IDRLnet [228]	https://github.com/idrl-lab/idrlnet
PINA [229]	https://github.com/mathLab/PINA
PhysicsNeMo (NVIDIA)	https://developer.nvidia.com/physicsnemo
NeuralPDE [230]	https://github.com/SciML/NeuralPDE.jl
TensorDiffEq [231]	https://github.com/tensordiffeq
ADCME [232]	https://github.com/kailaix/ADCME.jl
jinns [233]	https://gitlab.com/mia_jinns/jinns
PinnDE [234]	https://github.com/JB55Matthews/PinnDE
PINNacle [235]	https://github.com/i207M/PINNacle
PDEBench [236]	https://github.com/pdebench/PDEBench

4.3 Tools and Libraries

There exist several tools and libraries that can provide leverage and foundations for the development of PINN-based frameworks. Here we present some options. It is worth noting that most of these options prioritize Python, since it is a prevalent language in ML applications [1]. The repositories or websites related to the libraries mentioned below can be found in Table 4.

ML libraries are commonly utilized for Physics-Informed Neural Networks (PINNs). PyTorch [223] and TensorFlow [224] dominate academic research due to their native support for automatic differentiation, tensor operations, and scalability. JAX [225] is also a popular option. Similarly, PaddlePaddle [226] is an emerging alternative, particularly in industry-focused applications.

Beyond these general-purpose frameworks, domain-specific libraries provide high-level abstractions for physics-informed machine learning and PINNs, as well as a combination of useful tools or frameworks. DeepXDE [11] is such a library. It includes some PINNs variations, physics-informed DeepONets [101], and multi-fidelity PINNs [21]. It supports different backends. SciAnn [227] is a Keras/Tensorflow wrapper for PINN implementations. IDRLnet [228] is built upon PyTorch for PINNs, including functionalities for solving noisy inverse problems, variational minimization, and integral differential equations. PINA [229] is another PyTorch-based library, offering a user-friendly API specifically designed for scientific machine learning (Sci-ML) and PINN implementations that allows for flexibility regarding architecture and functions. TensorDiffEq [231] is a package built on TensorFlow for PINN-based solvers.

NVIDIA PhysicsNeMo (previously named Modulus) is the current iteration of what started as SimNet [216]. It is an open-source, free-to-use python framework and toolkit that offers multiple tools and algorithms for physics-based AI models and simulations, including PINNs.

For alternative options, NeuralPDE [230] and ADCME [232] are similar physics-informed ML oriented libraries, which are written in Julia.

Additionally, newer libraries seem to be emerging, such as jinns [233], which relies on the JAX ecosystem, and PinnDE [234], which is currently built for Tensorflow and JAX.

On a different note, PINNacle [235], is a Python-based (PyTorch and DeepXDE backends) benchmarking tool for PINNs. PDEBench [236] is a more general benchmark suite for Sci-ML that includes PINNs as one of the existing baseline surrogate models.

4.4 Future Considerations

As research and advances in continue to proceed, we can expect new, original concepts surrounding PiML, PINNs and their applications. These include:

(1) **Further component improvements.** Regarding the components of PINNs, recent publications such as adaptive [22, 23] or specialized activation functions [24, 161], meta-learners [168], and optimizer extensions [163, 166], demonstrate that PINNs have noticeable differences compared to artificial neural networks and benefit from different approaches. Based on these trends and observations in the literature, we can expect further advancements in regard to their components and enhancing strategies. Probable future directions include novel smooth and/or adaptive activation functions, PINNs oriented optimizers, sampling and weighting strategies, resulting in a toolkit of components optimized specifically for physics-informed learning, potentially diverging further from standard deep learning practices.

(2) **More diverse architectures.** Standard PINNs are MLP-based, and can often provide good results. They are also easy to implement, and improvements, such as the improved MLP [7] or Fourier feature embeddings [123], are available. However, as PINNs evolve and handle complex problems we can expect a shift towards more targeted, feature- or physics- aware choices of architectures. In the cases of convolutional architectures, their spatial awareness can enhance the framework, and such approaches can be significantly useful for materials science (by extent, engineering and physics) when training data are images or grids - or image/grid-like (e.g. [42, 125],). Similarly, GNNs offer a powerful alternative, providing the functionalities needed for spatial reasoning and generalization that mesh-free MLPs lack, as well various levels of 'mesh dependency'. For instance, MeshGraphNets [148] generalize to unseen topologies, and Graph Network-based Simulators (GNS) [149] scale to systems orders of magnitude larger than their training data, both demonstrating advanced physical consistency. Such notable examples highlight the importance of architecture schemes regarding the successful integration of physical knowledge. To this end, novel equivariant schemes are also great candidates for future PINNs and their applications. A transition towards more diverse, feature-aware architectures, is expected for future PINNs applications.

(3) **More specialized PINNs.** PINN configurations can vary significantly. However, recent advances introduce highly tailored architectures where the network design is tightly coupled with the underlying physics. Cases like PICNN [125] or enriched-PINN [50], FP-M/ FP-P [208] and RheologyNet [33], or tailored multi-networks for multiphysics cases [154] could be considered such PINNs. Their configurations are significantly intertwined with the characteristics of the problem they solve, thus they cannot easily transfer to alternative problems. A possible future trend is the emergence of more specialized PINNs, creating a distribution of PINN solutions per type of problem. We recognize that PINN variants, enhancements, and distinct configurations absolutely do this to an extent, but

they do not translate to specialization to such a degree. For example, VPINNs can be applied to several types of problems that are easier to solve when the PDE is in the variational (weak form). In contrast, specialized PINNs could be beneficial for complex industrial applications, facilitating the development of highly customized tools.

(4) **Industrial applications.** It is logical to expect the application of PINNs on a more industrial level. While early works often focused on simple cases or examples, the complexity of applications continues to evolve. Several cases visited in Section 3, demonstrate that PINNs have been applied to multiphysics scenarios, or more complex problems, with promising results. The more PINNs are investigated and improved, the more complex and realistic their applications can be.

(5) **Novel usages.** PINNs are already quite flexible, being able to be applied in both forward and inverse settings, or to be used as surrogates. However, there are many insightful ways to utilize PINNs. As seen in some of the examples above, PINNs have been applied as real-time field solvers [59] for optimized materials design, as nondestructive evaluation techniques for defect identification [55, 56], as a novel method for homogenization [61], as well as for the improvement of physical models [196]. Examples like these open the door for novel and diverse applications of the PINN framework. PINNs can also be used as components of larger frameworks of combined strategies, as seen in [193].

(6) **PiML method evolution.** While this review focuses on PINNs, the fusion of physics and governing laws with other advanced AI paradigms beyond the PINN framework constitutes an influential future direction. As highlighted in recent methodological reviews, generative models and reinforcement learning are increasingly explored for applications like topology optimization [237]. This aligns with a broader vision for computational mechanics that leverages these techniques for more integrated and autonomous AI frameworks [238]. Thus, a key research frontier is the development of novel physics-informed generative and reinforcement learning approaches, as well as physics-informed neural operators, specifically tailored for complex engineering tasks such as autonomous structural design, risk assessment, and materials discovery.

5 Conclusion

In this work, an overview of PINNs and their applications in materials science is presented. PINNs are a promising tool, a deep learning scheme combined with the representation offered by physical models. Since PINNs' configuration is quite flexible and usually tailored to the problem at hand, it is important to highlight different options and strategies for a sound, robust approach.

With regard to strategies and enhancements, discussion is provided regarding the different building blocks of PINNs. We discuss both the deep learning components (neural network specifics, e.g. architecture and activation functions), as well as the physics-related components such as the physics-informed loss function and adaptive weighting strategies, sampling strategies in space and time, boundary condition enforcement. In addition, diverse PINNs variants and extensions are provided.

As the authors of this work, we realize that we have not included every PINNs-adjacent publication under the scope of materials science. The main purpose of this survey is to present several applications in an organized manner, aiming to emphasize the variety of feasible PINNs-based solutions that are available in the literature. For our approach, we consider heat transfer one of the most vital phenomena in the context of materials science, thus we present several cases of PINNs applications for heat transfer problems, with respect to the governing mechanism and the nature of each case. Specifically, we discuss heat conduction, advection/convection - diffusion, radiative transfer, as well as phase change problems including the free boundary Stefan problem. In addition, we present heat-related multiphysics and realistic-industrial cases. Subsequently, we dedicate a section to materials modeling, properties, and design, where we present examples from the literature regarding several diverse settings. We also consider materials durability and include studies and applications oriented toward defect detection and analysis, fatigue life estimation, and fracture analysis or crack propagation, while distinct PINN applications for stress analysis are also presented. We conclude our investigation with instances of PINNs applications for porous media and multi-phase materials.

The final part of this survey presents extended discussions about several vital aspects of the PINN framework, including advantages, disadvantages, challenges and detailed future directions. In addition, an informative list of PINN-related libraries and tools is included.

Author Contributions A.M.: conceptualization, analysis, research and execution. I.P.: conceptualization, editing, supervision. I.G.: editing, supervision. S.V. and I.K.: funding acquisition.

Funding This work was supported by the Horizon Europe Framework Programme and EC-funded project DiMAT, EU under grant agreement No 101091496.

Data Availability No datasets were generated or analysed during the current study.

Code Availability Not applicable.

Declarations

Ethics Approval and Consent to Participate Not applicable.

Consent for Publication Not applicable.

Competing Interests The authors declare no competing interests.

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